

SURVIVAL
at ALTITUDE
FOR
HEAVY AND
VERY HEAVY
BOMBER CREWS

2ND A.F. ALTITUDE
INDOCTRINATION UNIT
L.A.A.F. LINCOLN, NEBR.
18 OCT 1944

SURVIVAL
at ALTITUDE
FOR
HEAVY AND
VERY HEAVY
BOMBER CREWS

2ND A.F. ALTITUDE
INDOCTRINATION UNIT
L.A.A.F. LINCOLN, NEBR.
18 OCT 1944

The Project Gutenberg eBook of Survival at Altitude for Heavy and Very Heavy Bomber Crews

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

Title: Survival at Altitude for Heavy and Very Heavy Bomber Crews

Creator: 2nd. Altitude Indoctrination Unit United States. Army Air Forces. Air Force

Release date: November 24, 2021 [eBook #66809]

Most recently updated: October 18, 2024

Language: English

Original publication: United States: 2nd Air Force Altitude Indoctrination Unit.

Other information and formats: www.gutenberg.org/ebooks/66809

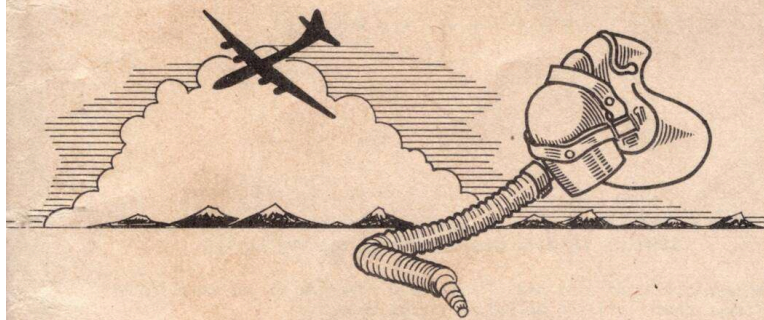
Credits: Kenneth R. Black, Sue Clark and Roger Frank

*** START OF THE PROJECT GUTENBERG EBOOK SURVIVAL AT
ALTITUDE FOR HEAVY AND VERY HEAVY BOMBER CREWS ***

Typographical errors (such as "COMAT" appearing for "COMBAT" in the title) and punctuation inconsistencies have been retained. The goal has been to provide a historically accurate representation of the original 1944 publication.

RESTRICTED

SURVIVAL *at* **ALTITUDE**
FOR
HEAVY *and* **VERY HEAVY**
BOMBER CREWS



PREPARED BY
2ND A.F. ALTITUDE INDOCTRINATION UNIT
L.A.A.F. LINCOLN, NEBR.

PROPERTY OF.....

RESTRICTED

NOTES ON THE USE OF OXYGEN EQUIPMENT
IN THE B-17, B-24, & B-29 FOR COMAT CREWS

* * * * *

THIS PUBLICATION MAY BE USED BY PERSONNEL RENDERING SERVICE TO THE
UNITED STATES OR ITS ALLIES

Paragraph 5.d. of Army Regulation 380-5 relative to the handling of "restricted" printed matter is quoted below.

"d. Dissemination of restricted matter.--The information contained in restricted documents and the essential characteristics of restricted material may be given to any person known to be in the service of the United States and to persons of undoubted loyalty and discretion who are cooperating in Government work, but will not be communicated to the public or to the press except by authorized military public relations agencies."

RESTRICTED

FOREWORD

These notes were originally intended as a source of general reference concerning OXYGEN EQUIPMENT AND ITS USE by members of crews of the B-17 and B-24 heavy bomber. With the advent of the B-29 as a combat bomber, the section on THE B-29 AT HIGH ALTITUDE (Page) was added to supply certain specialized information concerning this ship. Every B-29 crew member should be familiar with the INTRODUCTION and the section on THE DEMAND OXYGEN SYSTEM as well as this specialized information. The subject matter is presented from a PRACTICAL point of view, based upon actual experiences encountered on altitude missions. Detailed information on oxygen equipment may be found in the following TECHNICAL ORDERS.

03-50-1	Use of Oxygen and Oxygen Equipment.
03-50A-5	Type A-12 Demand Oxygen Regulator (Pioneer).
03-50A-8	Type A-12 Demand Oxygen Regulator (Airco).
03-50B-1	Type A-10 Revised Oxygen Mask.
03-50B-6	Type A-14 Demand Oxygen Mask.
03-50C-3	Low Pressure Oxygen Cylinders.
03-50D-1	Oxygen Pressure Signal Assembly.
03-50D-2	Oxygen Flow Indicator Type A-1.
03-50D-4	Oxygen Flow Indicator Type A-3.
03-50D-5	Oxygen Pressure Gage Type K-1.

INDEX

Auto-Mix	17
Ceiling, effect of oxygen on	3

Dangers with oxygen	<u>50</u>
Demand oxygen system	<u>7</u>
Demand type masks	<u>8</u>
Duration of oxygen supply	
B-17	<u>38</u>
B-24	<u>44</u>
Method of computing	<u>40</u>
Emergency valve	<u>19</u>
Explosive Decompression	<u>61</u>
Flow indicator	<u>32</u>
Hazards of oxygen-lack	<u>1</u>
Leaks in oxygen lines	<u>49</u>
Leaks in regulator	<u>11</u>
Mask, Demand	
Fitting to face	<u>9</u>
Technic, at altitude	<u>13</u>
Testing for leaks	<u>10</u>
Mask-Regulator connection	<u>16</u>
Night Vision	<u>2</u>
Oxygen cylinders	<u>37</u>
Oxygen deficiency at altitude	<u>3</u>
Oxygen installation, Servicing of	<u>47</u>
Oxygen discipline	<u>4</u>
Oxygen systems	
B-17	<u>36</u>
B-24	<u>43</u>
B-29	<u>57</u>
Oxygen systems, diagrams of	
B-17	<u>38</u>
B-24	<u>47A</u>
B-29	<u>60</u>
Oxygen equipment, check of	<u>34</u>

Portable oxygen equipment	<u>20</u>
Precautions with demand system	<u>34</u>
Passing out at altitude	<u>51</u>
Pressure Cabins, B-29	<u>56</u>
Regulator, Demand	
A-12	<u>17</u>
A-13	<u>20</u>
Regulator hose clip	<u>16</u>
Revival of passout victim	<u>51</u>
Servicing oxygen installations	<u>47</u>
Signal assembly	<u>32</u>
Sniff-tester	<u>11</u>
Testing for mask leak	<u>10</u>
The B-29 at High Altitude	<u>53</u>
Walk-around bottle	<u>20</u>
Walk-around bottle	
Uses of	<u>20</u>
Large, for engineer	<u>25</u>
When to use oxygen	<u>1</u>

INTRODUCTION

As a member of a heavy bomber team, in combat you will be living and fighting in a world where man has no business--from three to six miles above the environment for which nature designed you.

Successful and effective operation of combat crews at altitudes in excess of 15,000 feet requires that EVERY crew member be familiar with altitude problems. You must be so thoroughly familiar with your oxygen equipment that you will use it PROPERLY without giving it a thought. The time to acquire this experience is during operational training not after you have pulled several boners in combat.

Through long hours of tiring training you have developed superior judgment, ability to make decisions quickly, and efficiently in firing and in other duties. Most of your training has been confined to relatively low altitudes in a relatively "normal" environment. Now you are going to continue this training "upstairs". To maintain the superiority and efficiency you possess OXYGEN MUST BE USED above 10,000 feet. Above 18,000 feet oxygen must be used to maintain consciousness and life itself.

WHEN SHOULD FLYERS USE OXYGEN?

1. When wounded, especially with injuries accompanied by hemorrhage or shock.
2. Above 10,000 feet on day missions, both in training and combat.
3. From ground up on all night missions.
 - a. To prevent night blindness.
 - b. Night vision impaired early and to a marked degree at comparatively low altitude.

HAZARD OF OXYGEN-LACK

1. At 12,000 to 15,000 feet (several hours): fatigue, drowsiness, headache.
2. At 15,000 to 18,000 feet (½ to 1 hour):
 - False sense of well being
 - Overconfidence
 - Narrowing of field of attention
 - Faulty reasoning
 - Poor judgement
 - Loss of self-criticism
 - Clumsy
 - Blurring and double vision
 - Decreased hearing
 - Poor memory
 - May pass out
3. Above 18,000 feet:
 - Symptoms come on faster
 - Loss of muscle control
 - Loss of awareness of lapse of time
 - Loss of judgment and self-criticism
 - Loss of memory and ability to think
 - Loss of vision and hearing
 - Purposeless movements, repeated over and over
 - Emotional outbursts--hysteria, crying, laughing, anger
 - Loss of consciousness eventually:
 - at 26,000 feet 4 to 6 minutes
 - at 28,000 feet 2 to 4 minutes
 - at 30,000 feet 1 to 2 minutes
 - at 38,000 feet 30 seconds or less
 - The above varies with different individuals.
4. Loss of night vision.

Night vision is reduced to one-half at 12,000 feet without oxygen.
Breathing oxygen restores it to normal.

5. Breathing oxygen raises your ceiling.

ALTITUDE	%OXYGEN	EQUIVALENT
FEET	BREATHED	LEVEL (feet)
18,000	60	ground

26,000	85	ground
30,000	100	ground
33,000	100	8,000
38,000	100	12,000
44,000	100	22,000

REASONS FOR OXYGEN DEFICIENCY AT ALTITUDE

1. Inadequate supply for the mission. The low pressure oxygen system in the B-17 and B-24 is fully charged to a capacity of 450 pounds per square inch. This supply should last 7 to 9 hours, depending on activity of the crew members (Auto-Mix "ON").
2. Oxygen supply exhausted by leaks in oxygen lines or cylinders.
3. Flyer getting insufficient oxygen:
 - a. Attempts to do excessively hard work.
 - b. Mask leak: Improper fit; frozen exhaust valve; holes in mask or hose; growth of stubby beard.
 - c. Leak around gasket between mask-regulator connection.
 - d. Loose connections where supply hose connects to regulator.
 - e. Freezing of moist oxygen at extremely low temperature.
 - f. Obstruction of openings in regulator.
 - g. Hole in rubber diaphragm in regulator.
4. Breathing too deeply may produce a partial vacuum inside mask causing air to leak in from bottom or sides. **BREATHE NATURALLY.**
5. Don't expect to get oxygen from an empty walk-around bottle. If not fully charged, the bottle may last only a minute or two.
6. Improper use of walk-around equipment.

OXYGEN DISCIPLINE

It is mandatory that every member of a heavy combat crew be highly trained in oxygen discipline. It is urged that oxygen "drill" be practiced by ALL crew members for at least 15 minutes on routine training flights before altitude missions are ever undertaken. Practice using and re-filling the walk-around bottle at various stations. Practice changing masks (page 12) and become expert at using oxygen from emergency sources (page 27). Practice

reviving some crewmate who has supposedly passed out from oxygen lack. Practice going to your escape hatch on a walk-around bottle with parachute in place. Practice all emergencies that may arise some day in combat and when they arise they won't be emergencies.

The following responsibilities will fall upon every member of a combat crew. The airplane commander should satisfy himself that the following points are observed or provided for:

1. Fitting of the crew with masks, (page 9). Periodical check of masks and mask fit.
2. All crew members are familiar with hazards of oxygen-lack at altitude. (page 1).
3. See that all crew members are drilled in "precautions with the demand system." (page 19).
4. Be sure that ALL crew members realize the danger of mixing oil and oxygen. (page 50).
5. Be able to compute oxygen duration in different systems on ship (pages 39 & 46).
6. PRE-FLIGHT CHECK OF OXYGEN EQUIPMENT.
 - a. Pressure in different systems.
 - b. Be able to charge oxygen system if necessary. (page 47).
 - c. Check regulators for proper setting (auto-mix), properly working diaphragm, hose-regulator connection, emergency valve OFF.
 - d. Check EACH station for possible oxygen deficiency at altitude, (page 3).
 - e. Check portable oxygen equipment, regulator diaphragm, and oxygen supply in bottles, see that regulators are working properly.
 - f. Check all recharger hose valves for sluggish or sticking valves (page 20).
7. Know how to re-distribute crew members to stations where oxygen is available if part of supply is lost. (pages 41 & 45).
8. Someone to be able to find and repair a leak in oxygen lines. (page 49).
9. Frequently check condition of crew members during flight. See that one crew member checks on a fellow crewmate when practicable. Check every 10-15 minutes above 20,000 feet. Co-Pilot can help in this regard.

10. All crew members understand principles of reviving a crew member unconscious from oxygen lack. (page 51).
11. Report to crew chief and personal equipment officer on landing any and all faulty oxygen equipment. Make sure repairs to such equipment have been made before next flight.

THE DEMAND OXYGEN SYSTEM

The "demand" oxygen system in your ship insures a fully adequate oxygen supply except under the most extreme conditions. It was developed to supply you, AUTOMATICALLY, as much oxygen as you need when you need it--at all altitudes and under all conditions.

The DEMAND OXYGEN SYSTEM consists of the following:

1. Demand type mask.
2. Mask to regulator tubing.
3. Demand type regulator (A-12).
4. Portable recharging hose (for filling walk-around bottles)
5. Portable equipment:
 - a. A-4 Cylinder with A-13 demand regulator.
 - b. D-2 cylinder with A-13 regulator.
6. A panel containing:
 - a flow indicator
 - an oxygen pressure gauge
7. Cylinders containing breathing oxygen.

DEMAND TYPE MASKS

Styles: A-10A and A-14

The A-10 mask is obsolete.

The A-10 revised mask is obsolete.

The A-14 mask is standard issue for staging and combat areas, and operational or replacement training schools.

The A-10-A mask is used in training where sufficient A-14 masks are not available.

Both the A-10-A and the A-14 masks are suspended from the summer and winter flying helmets.

NOTE: It is important that AERIAL GUNNERS obtain a properly fitting mask, fit to both summer and winter helmets at the first station available. Because of the extreme cold to which gunners are subjected under combat conditions, you MUST learn to remove and replace the mask quickly and accurately WEARING HEAVY GLOVES. It is not easy to manipulate a mask or unjam a .50 cal. gun with numb fingers in heavy flying gloves. PRACTICE these things on every training flight with gloves ON.

FITTING THE MASK TO THE FACE -- There are two considerations in obtaining a proper fit:

1. An AIR-TIGHT SEAL of mask to face.
2. Comfort of fit without undue pressure.

The A-10-A and A-14 masks come in three sizes. Roughly, 60% of flyers will require the STANDARD (medium) size mask, 30% the LARGE mask, and the remaining 10% the SMALLER sizes. General instructions concerning fitting are.

1. Suspension points on both summer and winter helmets must be determined INDIVIDUALLY on each flyer. This is a job for your squadron personal equipment officer or the group aviation physiologist or his trained assistant.
2. It is important that the mask be suspended high on the face, sealing to the boney portion of the nose and cheek bones and NOT the soft tissues. Otherwise you'll get a leak or a "clothespin on the nose" effect from pressure of the nose wire.

NOTE:

- a. Be sure the regulator hose clip or the clip on the walk-around bottle is clipped high enough on your clothing to remove all tension from mask hose--otherwise the mask will be pulled down on the face and will leak outside air.
- b. Above 25,000 feet with the temperature 20 degrees and lower, the tissues of the face contract and the rubber mask loses its pliability. Under these conditions, the mask tends to fall away from the face

and it becomes necessary to SHORTEN THE UPPER SUSPENSION STRAPS. On returning to warmer altitudes, these must be loosened again to avoid discomfort from a too tightly fitting mask. (Applies chiefly to gunners).

3. Excessive beard stubble will hold the mask away from the face and cause leak around the jaws.
4. See page 35 for care of mask.
5. TESTING FOR LEAKS -- Holding the thumb over the end of the mask tube (rapid disconnect) and inhaling GENTLY should cause the mask to collapse on the face. The most common place for a leak is alongside the bridge of the nose. The actual site of leak is determined by putting finger-point pressure beside the nose and over the cheek bones, one point at a time. The nose wire is then re-adjusted to cause pressure at this point. Be sure the mask is high enough on the face and on the bridge of the nose. If the seal is still below the boney portion of the nose try the next largest size mask.
6. "SNIFF-TESTER" -- An effective "sniff-tester" for detecting small leaks can be obtained as follows: Obtain one of the commercial nasal inhalers - Benzedrine, Vick's, Penetro (or metal tube of comparable size). Cut off the end ordinarily inserted into the nose flush with the barrel of the container, and remove contents. Fit barrel with an airtight cork stopper. Straighten a paper clip or take a piece of wire of similar gage, cut to 2½ inch length, and drive one end firmly into lower side of cork stopper. Fashion other end of wire into a 1/4 inch fish hook bend. Make a ball of cotton or tightly wrapped gauze, ½ inch in diameter, and attach to bent end of wire. Soak the cotton or gauze, in oil of peppermint or oil of cloves (obtained from squadron dentist or flight surgeon) and fit cork stopper, carrying the "sniff-tester" tightly into metal container. To "sniff-test", remove the stopper and hold the cotton plug on both sides of the nose at the top of the mask (eyes closed). Inhale gently with the mask hose blocked. A leak is present if the contents of the "sniff-tester" is detected. Rough as this test seems, it is very sensitive.

NOTE: The "sniff-tester" is also effective in testing the DEMAND REGULATOR for leaks. Put on mask. Fit snugly to face. Plug into regulator hose. Turn Auto-Mix (Air) Lever OFF. Hold sniff-tester close

to regulator diaphragm and breathe normally several times. If the contents of the sniff-tester" are detected, the regulator has a leak.

TWO TYPES OF "SNIFF-TESTERS"

- A. Cotton pledget soaked with oil of peppermint.
- B. Benzedrine inhaler containing blotter soaked with oil of peppermint.

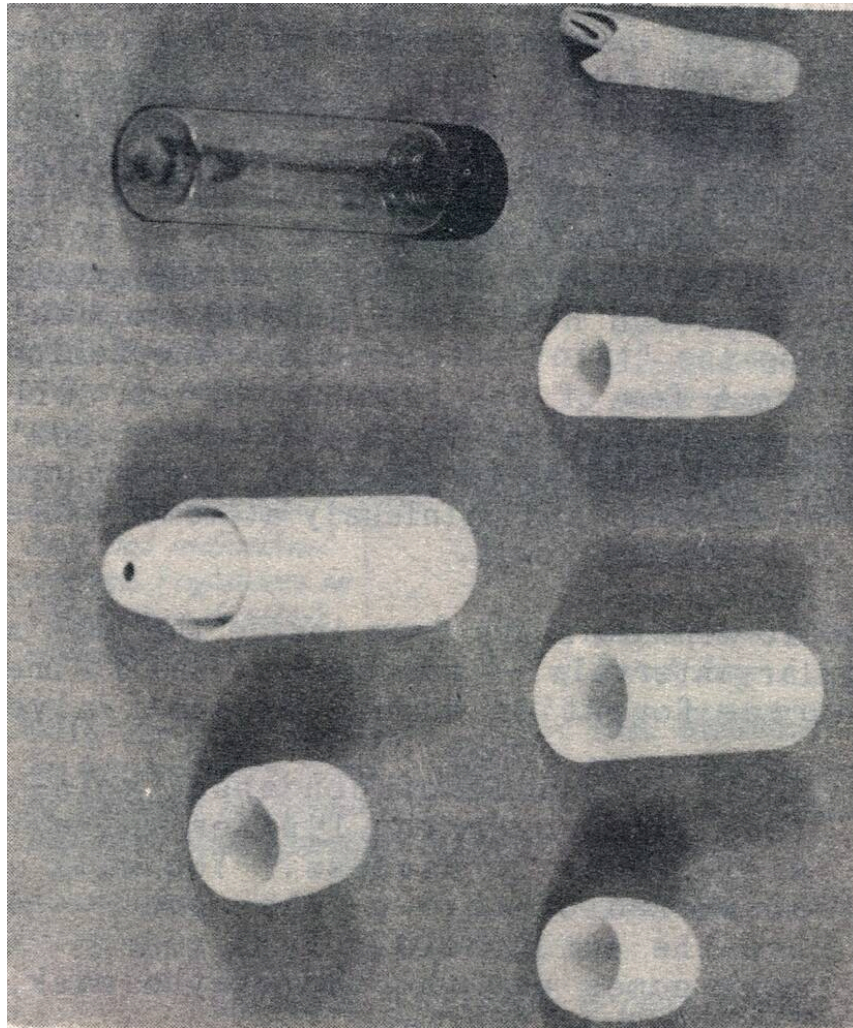


Fig. 1

REMOVING MASK AT ALTITUDE -- There are many reasons why your oxygen mask will be removed above 20,000 feet. This has been responsible for countless "accidents" in heavy bomber crews IN advanced training and in combat. **LEARN THE PROPER TECHNIC NOW,** wearing heavy

gloves. You can take your oxygen mask off at 30,000 feet and be as safe as you are on the ground, PROVIDING YOU HOLD YOUR BREATH and don't breathe the "thin" outside air. The following are a few of the reasons why you will remove your mask at high altitude. DON'T REMOVE YOUR GLOVES, for freezing occurs almost instantaneously and painlessly at temperatures of 30 below and lower.

1. To wipe moisture from inside the mask at regular intervals -- for comfort and to prevent ice formation around exhaust valve.
NOTE: A puddle of condensed moisture lying over the flutter valve will render it useless. You will be warned of this; when you breathe out, the air from your mask comes out the top of the mask, fogging your goggles.
2. To blow your nose. To help clear your ears on descent by holding nose and blowing (if necessary). To vomit if you get sick.
3. To take nourishment and hot drinks on missions.
4. To change masks.

TECHNIC -- NEVER breathe when mask is off face or is loosened from helmet. PRACTICE THIS with a mirror: (See Fig. 2,3 & 4)

1. Hold mask to face firmly with left hand.
2. Disengage mask clip from helmet with right hand.
3. Holding mask snugly in position with left hand, take 3 or 4 deep breaths of oxygen.
4. Remove mask with left hand, HOLDING BREATH. The right hand is free to service mask, to blow nose or to administer food or drink. Hold mask in left hand.
5. When you have to breathe, reapply mask to face making sure it is held snugly along sides of nose with index and middle finger of left hand, one on each side. of nose (hold mask in palm of hand with index and middle finger making a V, one on each side of the nose piece).
6. Take 4 or 5 deep breaths from the mask and repeat #4.
This procedure may be kept up indefinitely with complete safety, providing you DON'T BREATHE OUTSIDE AIR.

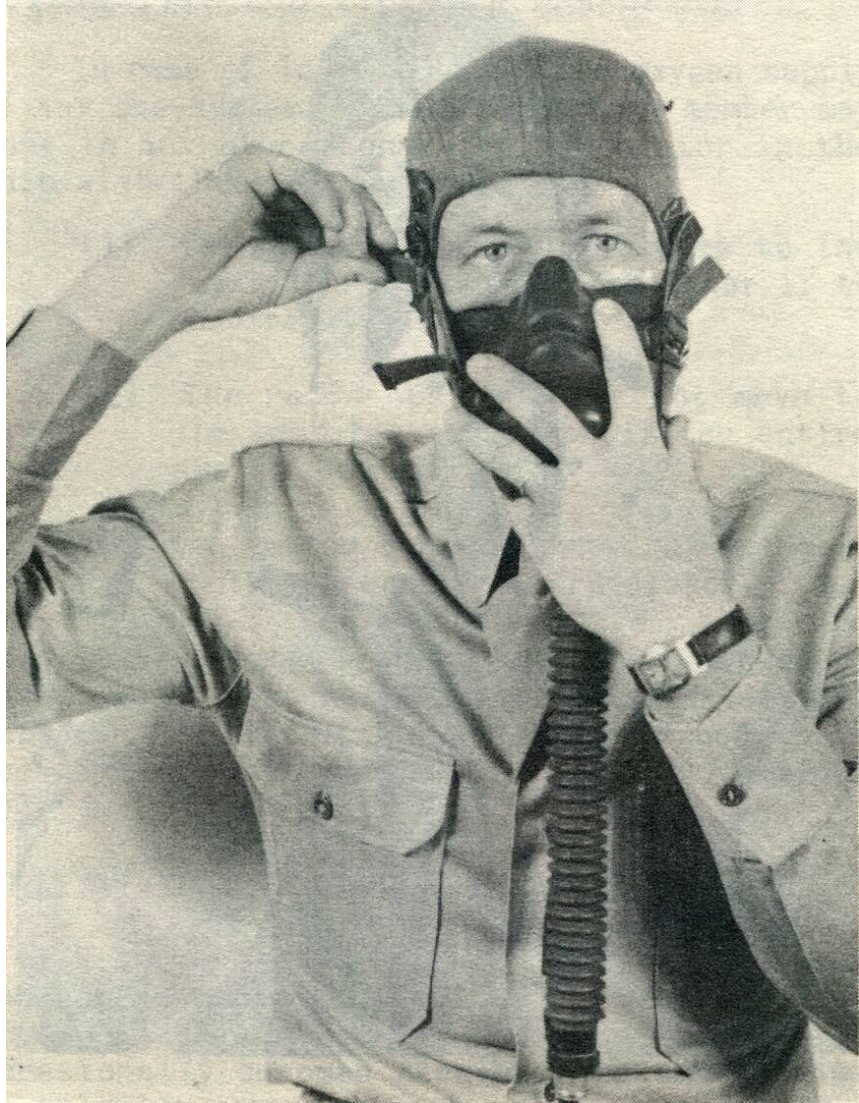


Fig. 2
MASK TECHNIC (1)



Fig. 3
MASK TECHNIC (2 & 3)



Fig. 4
MASK TECHNIC (4)

MASK TO REGULATOR CONNECTION (RAPID DISCONNECT)

IS THE RUBBER GASKET IN PLACE? Without the gasket an air-tight seal at this joint is impossible. Dangerous oxygen-lack will result above 24,000 feet. With a leak at this altitude serious mental and physical inefficiency will develop, even though you may not pass out. Above 28,000 feet you will quickly become punch drunk, and the pass-out will soon follow.

USE THE CLIP ON THE REGULATOR TUBING--It is there to remove tension on this connection which could (1) cause serious mask leak by pulling the mask down on the face, or (2) cause mask tubing to become disconnected. In combat these things would fail to attract your attention. Without warning you are first slap-happy and then drop unconscious.

THE RAPID DISCONNECT REQUIRES CONSTANT ATTENTION whether you are plugged into a fixed station or into a walk-around bottle. A pull of at least 12 pounds should be required to separate the male and female connection of the rapid disconnect. The security of the connection can be increased by spreading the groups at the tip of the male connection. (See diagram, pre-flight check, page 39).

CONNECTION OF HOSE TO REGULATOR IS NOT A SWIVEL JOIN-- Above 24,000 feet the collar must be turned up tightly to prevent a dangerous leak of air.

Make sure that the regulator hose is wired securely to the goose neck on the regulator.

THE DEMAND (A-12) REGULATOR

In principle the demand regulator is a diaphragm-operated flow valve which is opened by the user's inspiration and closes automatically when the suction ceases. This gives you as much oxygen as you ask for. All you do is breathe. A short breath draws a short squirt of oxygen, while deep, rapid breathing brings forth large shots in rapid succession.

The regulator can be set to conserve oxygen, or to give you 100% oxygen, regardless of the altitude you are flying. This is governed by the position of the "Auto-Mix" (AIR) lever.

AUTO-MIX (AIR) "ON" (or NORMAL OXYGEN) is the position for routine use. This saves oxygen.

1. Gives a mixture of air and oxygen:
 - 10,000 feet -- approx. 30% oxygen
 - 18,000 feet -- approx. 65% oxygen
 - 26,000 feet -- approx. 85% oxygen
 - 30,000 feet -- approx. 100% oxygen

Under the above conditions, so far as oxygen is concerned, you are at ground level.

2. Use oxygen from ground level on night missions to prevent night blindness.
3. Increased altitude increases oxygen supply AUTOMATICALLY.
4. Percentage of oxygen delivered keeps you at ground level efficiency.
5. Ship's oxygen supply, charged to 450 pounds lasts 10 men from 7 to 9 hours. (20,000 to 30,000 feet).

AUTO-MIX (AIR) "OFF (or 100% oxygen) -- used when ordered by first pilot, and on special occasions.

1. Gives 100% oxygen, on demand, regardless of altitude.
2. Used on special occasions:
 - a. To casualties suffering from:
 1. Loss of blood.
 2. Shock.
 3. Passing out from oxygen lack (revived).
 - b. Regulator not giving enough oxygen in "ON" position (blue finger nails-beginning to feel pleased with yourself).
 - c. To avoid gas poisoning if you land in a gassed area without gas masks.
 - d. To flush nitrogen gas from body fluids when going above 30,000 feet, helping to prevent bends.
3. Ship's oxygen supply, charged to 450 pounds, lasts 6 hours or less.

THE "EMERGENCY" VALVE IS DANGEROUS -- it should be used with extreme caution. A whole system can be depleted of oxygen in an hour or less if the "emergency" is opened only a quarter of a turn. Emergency "ON" changes regulator into a constant flow system and wastes precious oxygen. Use only to revive an unconscious crew member or to check line to make sure it is clear. Always make sure that "Emergency" is TIGHTLY TURNED OFF before ship leaves the ground.

NOTE:

Don't be confused by the difference in appearance of demand regulators manufactured by various companies (see Fig. 9). THEY ALL WORK THE SAME.

PORTABLE OXYGEN EQUIPMENT

A PORTABLE RECHARGER HOSE is available at every station in the ship. Every crew member should know their location so well that they can be located immediately, even on the darkest night. In addition, there is a long recharger hose for filling the ball turret oxygen cylinder. (B-17-F).

When drawing oxygen through a portable recharger the refiller valve must engage the filler nozzle **TIGHTLY**, otherwise oxygen leaks around the connection and is lost. The oxygen supply of an entire system has been wasted in this manner.

FREEZING or **STICKING** of refiller valve is not uncommon at extremely low temperatures. Always check this after refilling your walk-around. A stuck valve will quickly deplete the entire supply of oxygen in that system. If the valve is stuck plug in a walk-around bottle and leave it there.

USES OF THE WALK-AROUND BOTTLE -- (1) As an **ACCESSORY REGULATOR** -- the A-13 regulator on the walk-around bottle is a demand type regulator without the "auto-mix" feature. It gives 100% oxygen on inspiration regardless of the altitude. By connecting the walk-around bottle **FIRMLY** into the portable recharger valve, one has an extra source of oxygen at that particular station. As long as the oxygen gauge at the station registers oxygen pressure the user is assured a supply of pure oxygen ... his supply is coming directly from the system. Remember this when reviving a passed out crew mate in the waist in the vicinity of the ball turret--the long recharger hose used to charge the oxygen bottle on the ball turret can be plugged directly into a walk-around bottle attached to the unconscious crew member's mask.

If other long recharger hoses can be obtained from Tech Supply they will be extremely handy attached at the pilots station, in the bomb bay and to the recharger outlet at the right waist gunner's station. This would permit the engineer, who in flight is continuously filling his walk-around bottle, considerable liberty. He could cover most of the ship, staked out on a long recharger hose connected to his walk-around bottle. It would also provide a readily accessible source of oxygen to revive crew members who pick such out-of-the-way places to pass out as: the bomb bay cat-walk, the bomb bay doors, under the flight deck, or in some remote part of the waist.

One or two extension portable recharger hoses shown in photograph page 22 should be standard equipment in heavy bombers.

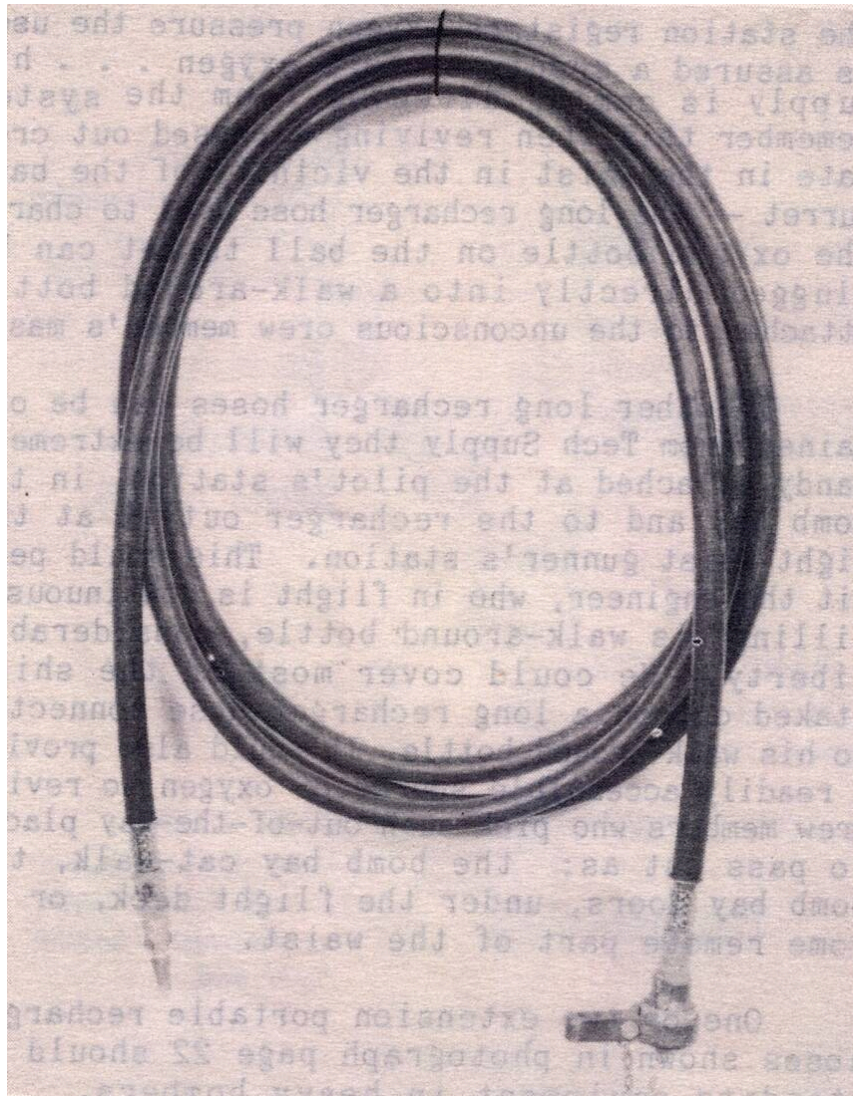


Fig. 5
EXTENSION RECHARGER HOSE

(2) SOURCE OF OXYGEN FOR BAIL-OUT -- Above 20,000 feet, breathing of oxygen is necessary to reach your particular escape hatch. After this spot is gained you won't need oxygen ... unless you start fiddling with the rip-cord too soon. Go to your exit on the walk-around bottle, fill your lungs deeply five or six times from the bottle, **HOLD YOUR BREATH**, and take the dive. Continue to hold your breath during the descent as long as possible. By the time your lungs are bursting for air

you'll be from 8 to 10,000 feet nearer the ground and won't need extra oxygen. Leave the mask on--it will protect your face. If the skies are filled with enemy planes it may be wise to continue the free fall. By now you can start breathing without danger of serious oxygen-lack. Any dimming of mental acuity would be of such short duration that you'd have plenty of time to pull the ripcord.

PRACTICE GOING TO YOUR ESCAPE HATCH ON A WALK-AROUND BOTTLE DURING OPERATIONAL TRAINING -- PRACTICE ON JUST HOW YOU WOULD CLEAR THE SHIP -- GO THROUGH ALL OF THE ROUTINE EXCEPT THE BAILOUT -- and that might not be time wasted, at least just once before the time comes when you might have to!

Above 30,000 feet you wouldn't want to open your 'chute even if you had plenty of oxygen. You'd freeze a hand or foot or both if you did. Under these conditions you'd free fall, holding your breath, as long as possible. Then, after three or four breaths of "thin" air, pull the rip cord. Your altitude should be in the vicinity of 10,000 to 12,000 feet below that when you abandoned ship, and the degree of oxygen-lack encountered here wouldn't be of dangerous consequence. You'll avoid freezing, and perhaps the gunfire of enemy interceptors.

(3) TO MOVE ABOUT THE SHIP For this the walk-around bottle is very useful ... and very tricky! The trouble is that the thing is good only as long as it gives oxygen. It will rarely, if ever, be fully charged so forget that 8 to 12 minute stuff. You can't fill your bottle any fuller than the pressure in the line you'll be drawing from. And the only time that will be fully charged is when the ship is on the ground, freshly charged with oxygen, ready for the take-off. When you need the walk-around bottle, three or four hours will have passed ... and the pressure in your filling line will be some fraction of the full charge. As a rule you'll do well to get the bottle half charged. And you're going to be moving about at altitude and needing 3 to 5 times as much oxygen as if you were sitting quietly in a corner. So plan on the walk-around supply lasting 1½ to 2 minutes ... then you won't be caught short! Refill the bottle frequently. Refill it every time you pass a portable recharger hose ... and there's one at every station. Failure to do this has caused more pass-outs in second and third phase training than any other single cause ... unless you include carelessness. And careless people don't

last long in the kind of work you're specializing in! There's a check-valve on the walk-around bottle, so you won't lose anything if you plug into a system that reads less than the gauge on the bottle.

NOTE TO ENGINEERS: Because of its limited supply, the standard walk-around bottle is next to useless for many of your jobs which must be done in flight on a walk-around bottle. The ENGINEER NEEDS A LARGER WALK-AROUND BOTTLE. One, having a supply 5 TIMES AS GREAT as the standard portable cylinder, can be obtained as follows:

Obtain a D-2 oxygen cylinder (stock No. 5500-344020 - class 03K) from Air Corps Supply. Remove the A-13 regulator assembly from a standard walk-around bottle. Remove the spud from one end of the D-2 cylinder and screw the A-13 regulator in tightly. Fill to 400 p.s.i. and leave overnight to determine if pressure is maintained or lost due to a leak. (A drop of 25 to 50 pounds will occur in the absence of a leak due to cooling of the oxygen which warmed up when the cylinder was charged).

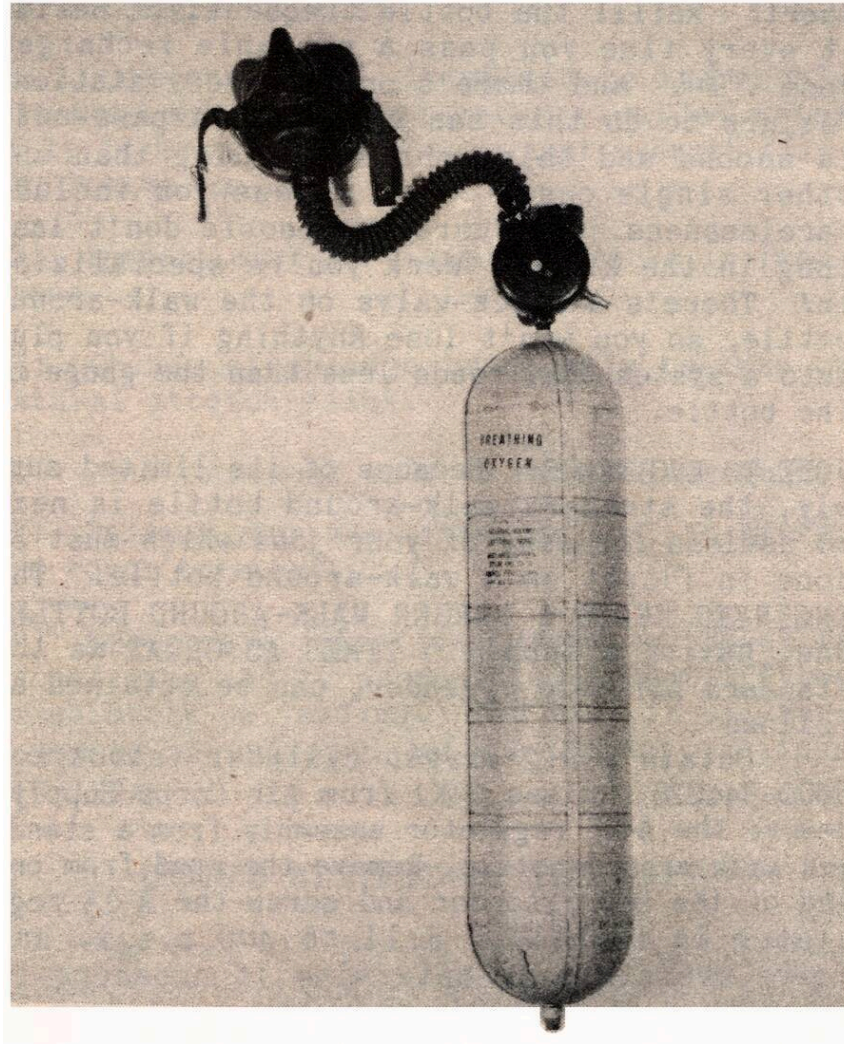


Fig. 6
D-2 WALK-AROUND BOTTLE

For leaks here or elsewhere in an oxygen system use the following anti-sieze and sealing compound on the pipe threads: Pioneer Antisieze No. 2., Class 96B, Stock no. 7500-050800. Never use a sealing compound which contains oil.

A satisfactory bag with shoulder strap for carrying the walk-around bottle when in use can be made from heavy twill or canvas by the parachute department.

EMERGENCY OXYGEN EQUIPMENT

In addition to your ship's regular oxygen supply, you will take emergency oxygen rations along on combat missions. If practical you will also have several extra masks on board. Know where they are stored.

1. Both of the low pressure walk-around bottles should be kept as full as possible-- you never know when you may need them. If full, the small walk-around bottle is good for only a few minutes, the large for 30 to 45 minutes, depending on your activity.

2. High pressure equipment.

- a. A-2 bottle with bag-mask. (Fig. 7)

In some theaters each combat crewman is issued this equipment. The high pressure bottle is fully charged to 1800 and will last approximately 45 minutes to one hour. It is used with the continuous flow mask.



Fig. 7
A-2 OXYGEN BOTTLE WITH BAG MASK

b. Bail-out bottle. (Fig. 8)

This small high pressure cylinder is a "last resort" source of emergency oxygen. It contains approximately 12 minutes supply. The old style bottle (H-1) is equipped with a pipe stem which is held between the teeth. A newer bail-out bottle (H-2) has a bayonet connection so it can be used with an A-14 mask with bail-out adapter (see fig. 8).

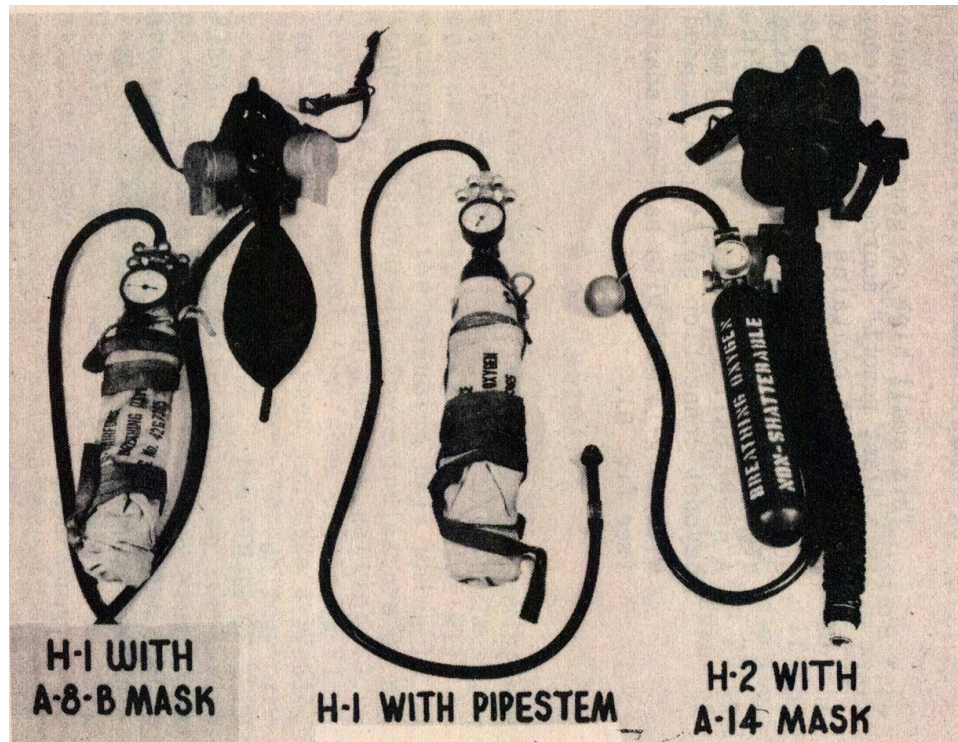


Fig. 8
BAIL OUT BOTTLES

PRESSURE GAGE

Your oxygen pressure gage registers from 450 pounds to zero. **WATCH YOUR PRESSURE GAGE!** Except under extreme emergency, don't run the pressure down under 50 pounds if you do there's danger of moisture getting into your refiller line. Then the next time you go to high altitude you'll have trouble with freezing in the oxygen line.

FLOW INDICATOR

Either the bouncing ball or blinking eye will be at each station, depending on the vintage of your ship. Only older ships will be equipped with the bouncing ball which is no longer manufactured. These indicators do not tell HOW MUCH oxygen you are getting ... they merely indicate that some oxygen is flowing through the system when you take a breath. They may be working O.K. even though your oxygen supply is inadequate. The COLOR OF YOUR FINGER NAILS is a better guide to the amount of oxygen you are getting, providing your hands are warm. If they become bluish above 20,000 feet check mask and connections for leak. Naturally, you won't remove your gloves to look at your finger nails if the temperature is 20 or 30 degrees below zero. If no leak is found, switch the "Auto-Mix" to the OFF position and note if things improve. INFORM YOUR PILOT IF AUTO MIX IF LEFT "OFF" and your mission is a long one.

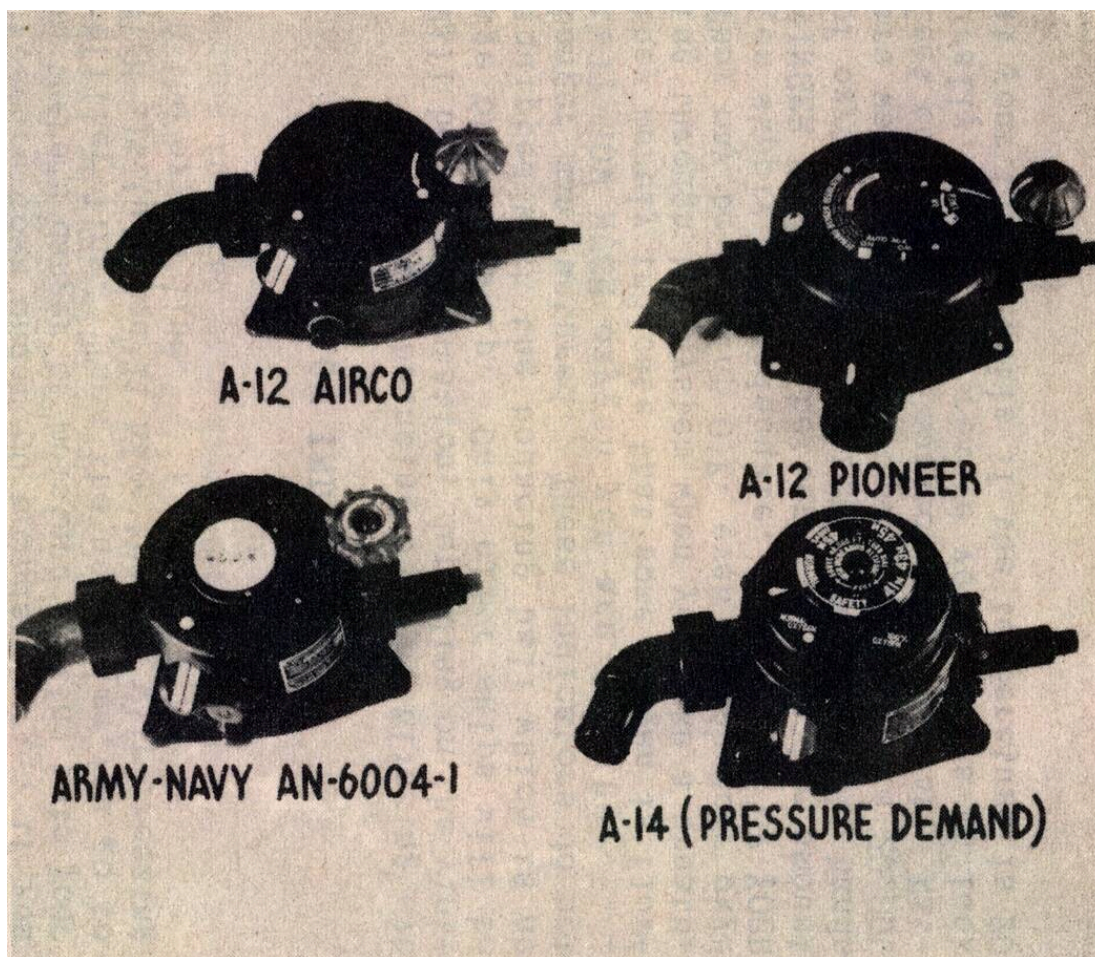


Fig. 9

PRE-FLIGHT CHECK

1-MASK

fit and check



washer



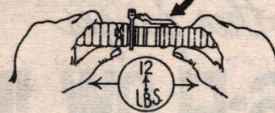
2-CONNECTION

12 lb. pull

*adjust
to fit snugly*



prongs

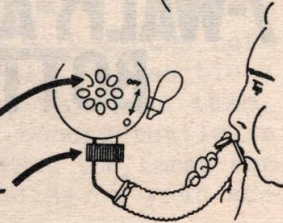


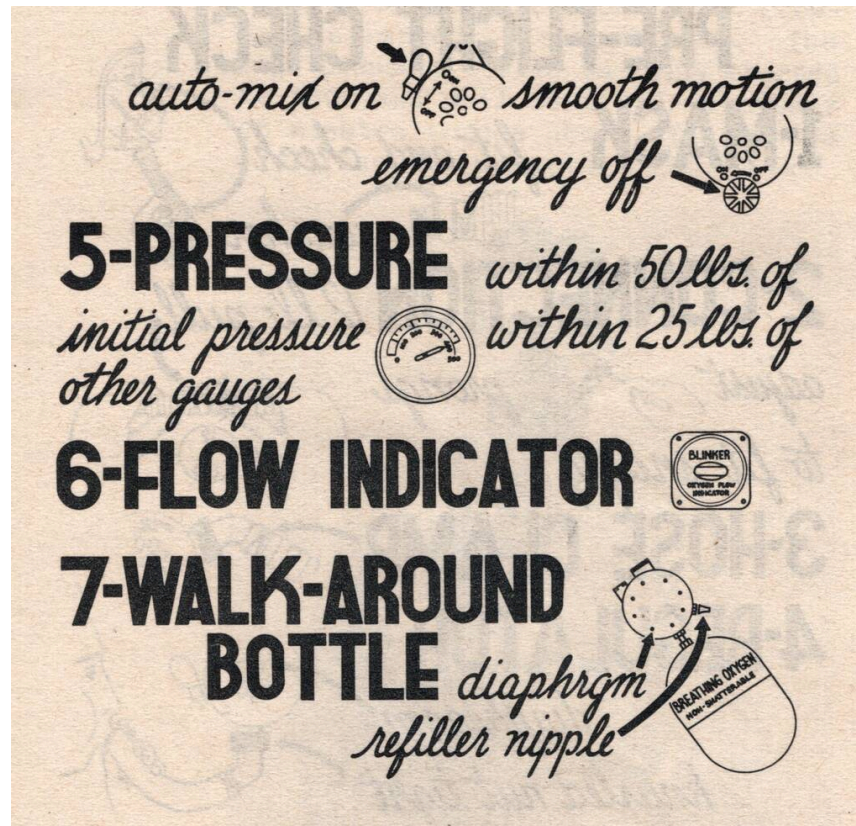
3-HOSE CLAMP



4-REGULATOR

*diaphragm
knurled nut tight*





IN THE AIR:

1. Check for leak by blockage and gently inhalation each time mask is removed and replaced.
2. Check oxygen flow indicator at regular intervals. Also, CHECK COLOR OF NAILS AT REGULAR INTERVALS -- if blue above 20,000 feet turn Auto-Mix to "OFF" position.
3. Check oxygen pressure gauge frequently.
4. If temperature is below freezing, manipulate mask to free it of ice at regular intervals. Wiping moisture from mask periodically will prevent freezing of mask.
5. Open EMERGENCY flow only when absolutely necessary ... notify pilot.
6. Over 28,000 feet switch Auto-Mix to "Off" position ... turn back to "ON" position below this altitude.
7. Re-fill walk-around bottle frequently when you are using it at altitude.
8. Avoid unnecessary exercise above 20,000 feet ... take your time!

9. Observe your fellow crew member at frequent intervals above 20,000 feet. You can recognize before he can if he's in trouble.

ON RETURN TO FIELD

1. Wipe mask dry. It should be thoroughly cleansed with soap and water (inside) after 10 to 14 hours use. Be sure that exhaust flutter valve is kept clean.
2. Inspect mask for leaks or cracks in face-piece.
3. Change strap adjustment only to take up natural stretch slack.
4. Lend your mask only in extreme emergency.
5. Protect your mask from hot sun, moisture, and rough treatment.
6. Report all defective oxygen equipment to ground crew and to personal equipment officer immediately on landing. See that it has been repaired or replaced before another altitude mission.

OXYGEN SYSTEMS IN THE B-17

The Army Air Forces use the low pressure system (maximum charge 450 pounds per square inch) in preference to the high pressure system (1800 pounds per square inch) for these reasons:

1. Cylinders do not explode when hit.
2. Do not "rocket" from moorings when hit.
3. Less chance for fire or flash burns.

A diagrammatic sketch of the FOUR different oxygen systems is shown on page 38. EVERY crew member should thoroughly familiarize himself with the location of each regulator and each portable recharger hose.

The four systems, arranged two on a side, offer great advantages in combat over a single system. It is possible to redistribute crew members whose system has been shot out and stay in formation. Everyone must know these alternate positions. (page 41 for B-17, page 45 for B-24).

OXYGEN CYLINDERS

NOTE: Based on combat experiences, continuous changes are being made in the oxygen systems and oxygen supply in heavy bombers. The following considerations apply only to the type of ship specified. Newer models and ships modified at staging areas may be different both in arrangement and number of oxygen cylinders. KNOW the oxygen supply for YOUR STATION IN YOUR SHIP.

Two types are found in the B-17. EIGHTEEN G-1 cylinders contain the main supply which is distributed through the four systems. All are filled through a single recharging line.

The G-1 cylinder has a volume of 2100 cubic inches and contains 30 cubic feet of oxygen when charged to a pressure of 450 pounds per square inch. In combat the cylinders are charged to 450 pounds or better (all cylinders have been tested to withstand 700 pounds pressure). Each cylinder is check-valved so that one of a bank of cylinders may be shot out with loss of no oxygen from the remaining intact cylinders. Between 20,000 and 30,000 feet, one G-1 cylinder charged to 450 pounds will last one man approximately 5 hours (Auto-Mix "on").

TURRETS:

The turrets on the earlier models (B-17F) are supplied with F-1 type cylinders having a capacity of approximately 14 cubic feet when fully charged to 450 pounds. Under these conditions a fully charged bottle lasts one man approximately 2 hours. (Note: these bottles are recharged from the ship's supply during flight, so it will not be possible to fully charge them after several hours of flight on oxygen. They will be no fuller than the pressure in the system from which they are charged--front left system for top turret; rear left system for ball turret).

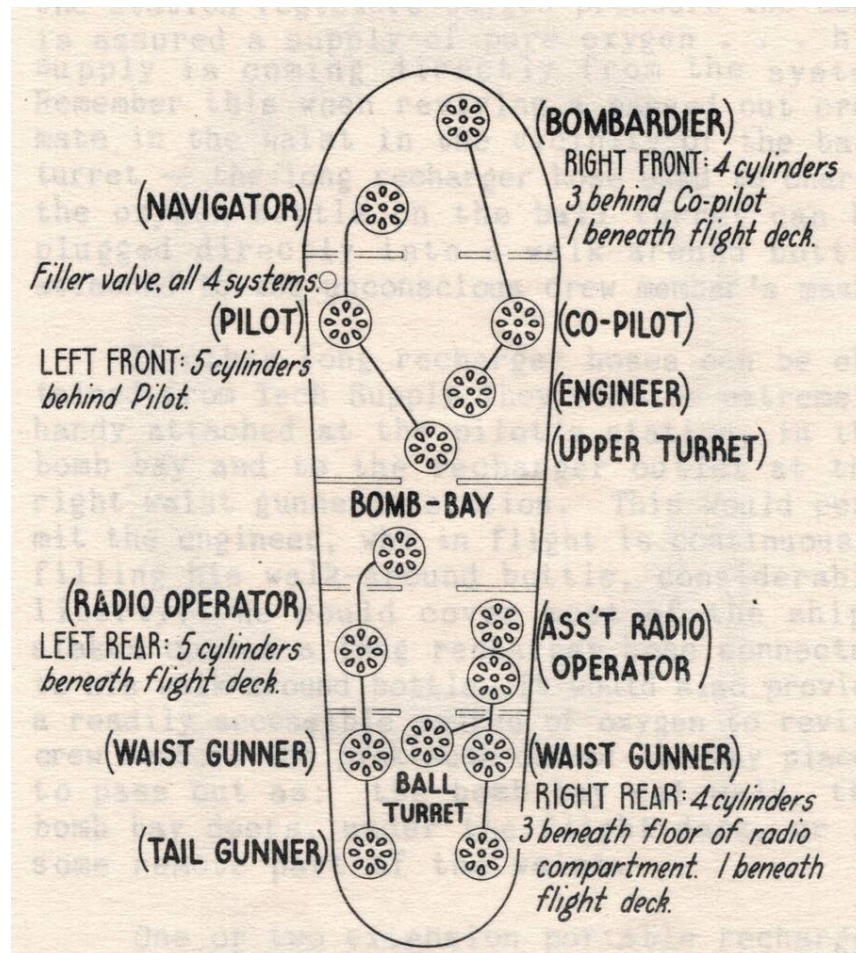


Fig. 10
OXYGEN SYSTEMS B-17G

B-17G models are modified so that the turrets are supplied oxygen directly through a flexible hose -- the top turret from the front left system -- the ball turret from the rear right system. The same changes may be found in the ball turret of the B-24.

COMPUTING OXYGEN DURATION

Several members of the crew must be able to calculate the duration of the oxygen supply. The following simple procedure will permit a fairly accurate estimate of each systems duration. The figures are based on altitudes between 20,000 and 30,000 feet where the majority of your combat flying will take place.

G-1 CYLINDERS -- full charge 450 pounds -- each cylinder equals 5 man hours.

FORMULA: 5 (man hours) x number of intact cylinders x actual pressure/400 divided by number of men using oxygen from system = DURATION in hours.

Examples:

FRONT LEFT SYSTEM - 5 cylinders -- full charge to 450 pounds = 25 man hours.

3 intact cylinders (2 shot out), pressure 200 pounds, pilot and navigator on system.

$$5 \times 3 \times 200/400 \times 1/2 = 3 \text{ plus hours.}$$

For 3 men:

$$5 \times 3 \times 200/400 \times 1/3 = 2 \text{ plus hours.}$$

FRONT RIGHT SYSTEM: 4 cylinders - full charge to 450 pounds = 20 man hours.

Example:

4 cylinders, pressure 160 pounds, bombardier, co-pilot, and engineer on system.

$$5 \times 4 \times 160/400 \times 1/3 = 2\frac{1}{2} \text{ plus hours}$$

LEFT REAR SYSTEM: 6 cylinders - full charge to 450 pounds 30 man hours.

Example:

6 cylinders, pressure 200 pounds, radio operator ball turret gunner, left waist gunner and tail gunner on system.

$$5 \times 6 \times 200/400 \times 1/4 = 3\frac{3}{4} \text{ hours}$$

RIGHT REAR SYSTEM: 3 cylinders - full charge to 450 pounds 15 man hours.

Example:

3 cylinders, pressure 300 pounds, two men on system.

$5 \times 3 \times 300/400 \times 1/2 = 5$ plus hours

Left rear system out - RO, ARO, AE and two AG's on right rear system.

$5 \times 3 \times 300/400 \times 1/5 = 2$ hours.

F-1 CYLINDER -- ball turret -- full charge to 400 pounds 2 plus hours

$2 \text{ (man hours)} \times 120/400 = 0.5$ plus hours (30 plus min)

ALTERNATIVE POSITIONS IN EMERGENCY (B-17G)

Left front system out:

Navigator on walk-around bottle plugged into Bombardier's recharger hose. Pilot uses Engineer's regulator hose from right front system.

Right front system out:

Bombardier on walk-around bottle plugged into Navigator's recharger hose. Engineer same on Pilot's recharger hose. Copilot on top turret regulator hose (hose too short in later models).

Left rear system out:

RO on one regulator from across aisle.

ARO on regulator right radio compartment.

AE on walk-around bottle on recharger hose, right radio compartment.

AG on waist regulator, right.

AG on tail regulator, right.

Right rear system out:

RO regular position.

ARO on bomb bay regulator.

AE on walk-around bottle plugged to ball turret recharger hose.

AG on waist regulator, left.

AG on tail regulator, left.

OXYGEN SYSTEM IN THE B-24

Like the B-17, the B-24 is equipped with a low pressure oxygen system. Location of the cylinders and distribution of the lines, however, is entirely different. The oxygen system varies somewhat in the different model ships

and in individual ships. In the B-24 D (beginning with Serial No. 42-40218) to the B-24 H, however, the system is essentially as described below. Some of the minor variations will be noted, including changes to the supply for the nose positions in late T models. FAMILIARIZE YOURSELF WITH THE OXYGEN SYSTEM IN YOUR OWN SHIP.

OXYGEN CYLINDERS:

There are 22 type G-1 cylinders arranged in eight systems (see diagram on page 47B). Sixteen cylinders are banked above the bomb-bay. The remaining six are buried in the floor of the rear fuselage compartment. There are also two smaller (type D2) cylinders attached to the chair of the top-turret gunner. These are not connected to the main system but must be charged from a portable recharger line. The main system is charged from a single filler valve located in the left side of the bomb bay. Each G-1 cylinder, when charged to 450 pounds per square inch, contains 30 cubic feet of oxygen, and will last one man approximately 5 hours between 20,000 and 30,000 feet (Auto-Mix "on").

DISTRIBUTION SYSTEMS:

The systems differ radically from those of the B-17. In models up through H to early J each individual draws from at least two or three cylinders. With the exception of the nose positions there is a maximum of two men on the same system. As shown in the diagrammatic sketch (page 47B), the pilot, co-pilot, side gunners, and tail gunner each have their own individual system of at least two G-1 bottles.

In the G, H and early J series the three regulators in the nose are supplied by a single system containing only three oxygen cylinders. (diagram page 47B). This is inadequate. The later J models have a modified system to the nose wherein the bomb bay - radio operator system is Tee'd into the supply to the nose positions. This brings to a total of six oxygen cylinders for the five stations. (diagram page 47A). Since the regulators at the radio operator and bomb bay stations are not used continuously during combat missions, this modification to the nose positions is adequate. Further, the connections in the lines leading to the nose are such that the nose positions are supplied individually as well as collectively so that loss of the supply to one system does not necessarily deplete the other systems. KNOW THE OXYGEN SYSTEM IN YOUR SHIP.

The bottom turret and camera tunnel regulators are supplied by the same system containing three cylinders.

ALTERNATIVE POSITIONS IN EMERGENCY:

In case of loss of the entire oxygen supply of any one system, the affected crew member may move to any other convenient regulator in the ship with these exceptions:

1. The nose gunner may not move to the navigator-bombardier regulator if the entire system is out.
2. The radio operator may not move to the bomb-bay regulator if the entire system is out.
3. The bottom turret gunner may not move to the camera tunnel regulator.

This is true because every station, with these three exceptions, has its own individual oxygen supply. The use of the walk-around bottle is the same as that described for the B-17. (Page 20).

DURATION OF OXYGEN SUPPLY:

With the present system in the B-24, each man has approximately 9 hours oxygen supply at 30,000 feet with the Auto-Mix on. However, it occasionally becomes necessary in emergencies to calculate the residual oxygen supply. This can be done by the simple formula given on page 24:

$$5 \text{ (man hours)} \times \text{no. of intact cylinders} \times \text{actual pressure}/400 \text{ divided by no. of men using the system.}$$

Application of this formula is much simpler in the B-24, since usually only one, and at the most two, men are using a single system.

VARIATIONS:

The exact location of the demand regulators in a ship is variable, depending upon the model, place of installation, etc. It is extremely important that you familiarize yourself with the location of the regulators in your ship, so that you can move from one to the other in darkness, in case of emergency.

Many of the B-24 H models have two extra regulators: one in the nose, and one in the tail. There is also variation in the location of the portable recharging hose for the top-turret bottles. This is located in the right side of the ship in most earlier models, but in the left side in later models. Fundamentally, however, the system in all ships commencing with No. 42-40218, is as described above. It is up to you to familiarize yourself with any minor differences in your ship.

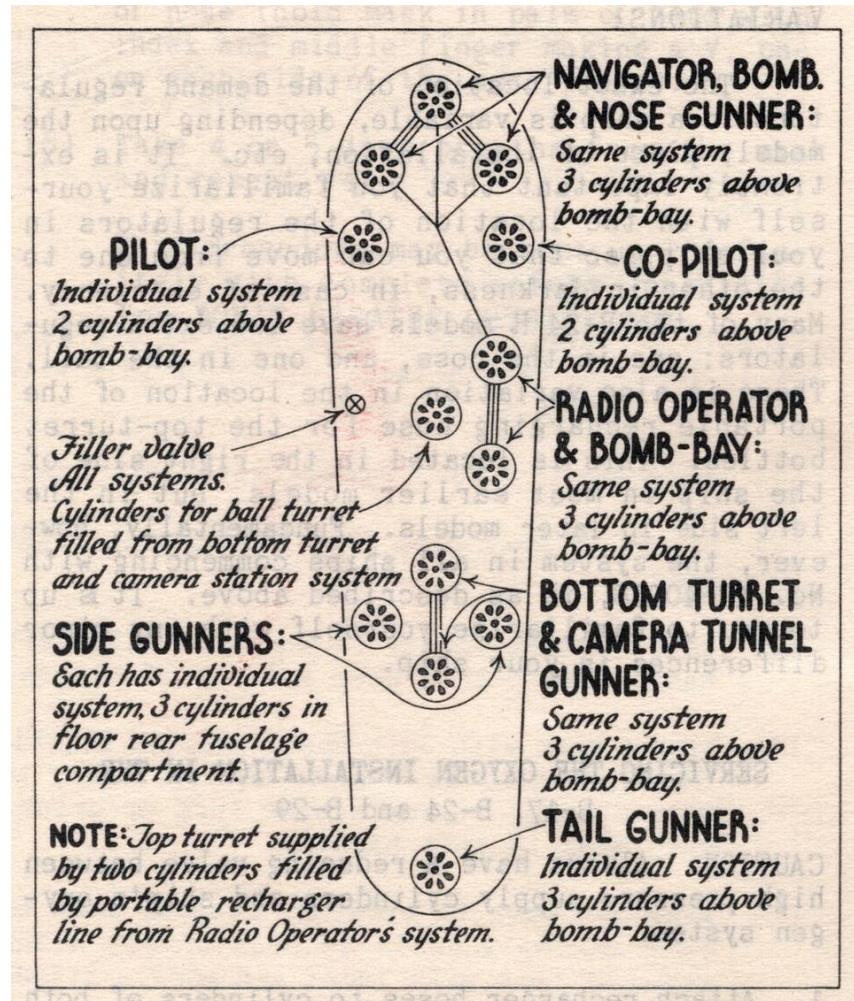


Fig. 11
OXYGEN SYSTEM B-24J MODIFIED

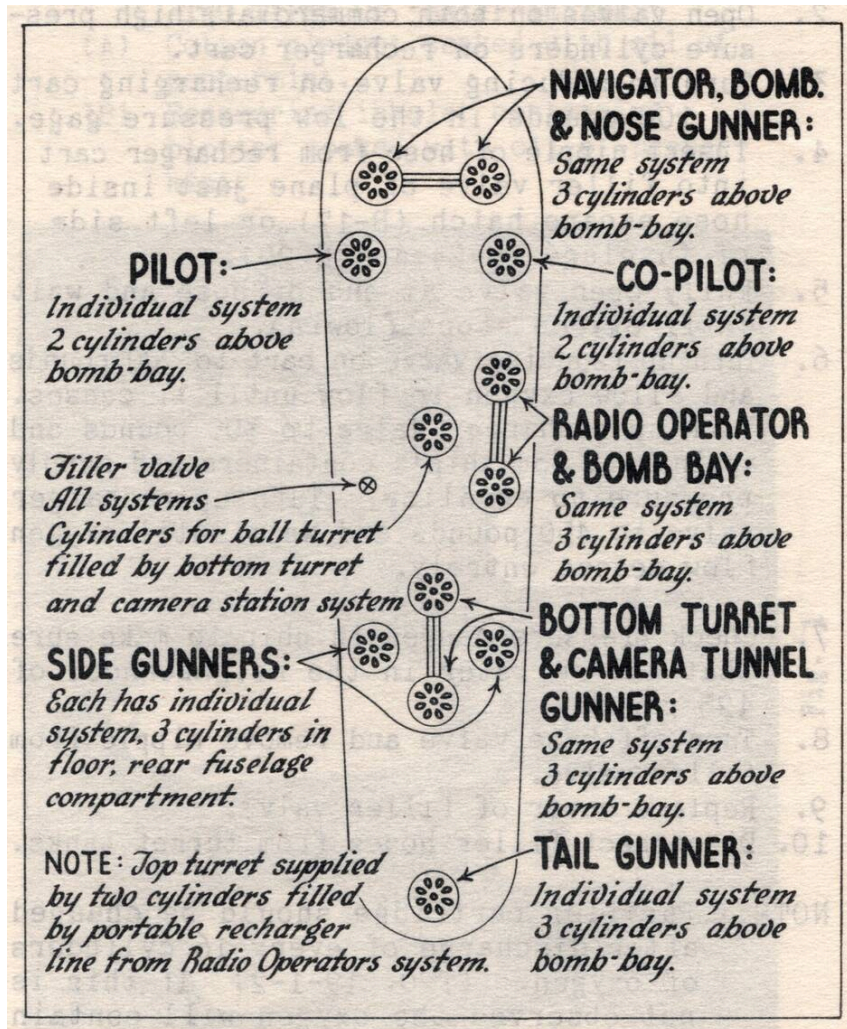


Fig. 12
OXYGEN SYSTEM B-24D to H

SERVICING THE OXYGEN INSTALLATION IN THE B-17, B-24 and B-29

CAUTION: Always have a reducing valve between high pressure supply cylinders and ship's oxygen system.

1. Attach recharger hoses to cylinders of both turrets in B-17 or B-24.
2. Open valves on both commercial, high pressure cylinders on recharger cart.
3. Turn in reducing valve on recharging cart to 100 pounds in the low pressure gage.

4. Insert nipple of hose from recharger cart into filler valve of plane just inside nose escape hatch (B-17) or left side of fuselage (B-24 and B-29).
5. Fully open valve at end of hose and wait until oxygen stops flowing.
6. Turn up reducing valve on cart to 200 pounds and allow oxygen to flow until it ceases. Turn up recharger valve to 300 pounds and again wait for ship's containers and supply pressure to equalize. Turn up recharger valve to 450 pounds and wait until oxygen flow ceases entirely.
7. Check pressure gauges in ship to make sure that all register in the neighborhood of 425 pounds.
8. Turn off hose valve and remove nipple from filler valve.
9. Replace cover of filler valve.
10. Disconnect filler hoses from turret tanks.

NOTE: Purifier cartridge should be changed after discharge of every 1.6 cylinders of oxygen. (T.O. 19-1-2) If this is not observed the oxygen will contain moisture and freezing in oxygen system will occur at high altitude.

After the oxygen in the tanks has had time to cool down, pressure should be in the neighborhood of 400 pounds.

The same procedure must be employed even if one or more of the systems is already fully charged since there is no way of equalizing pressure among the different manifolds. Do not attempt to use any system in which the pressure is less than 50 to 75 pounds per square inch. Under these conditions the oxygen flow is very weak. More dangerous is the possibility of moisture getting into the oxygen line through the main recharger valve. This will result in freezing in the line or one of the regulators on the next trip to high altitude.

LEAKS IN OXYGEN LINE

Leaks in oxygen lines can be detected by painting oxygen tubing with a paintbrush and solution of soap suds--soap bubbles appear at the site of leak.

Connection of oxygen tubing to nipples, elbows, tees, and regulators are made air-tight by applying an anti-sieze, sealing compound to the pipe threads (see note, page 27). DON'T use anything containing OIL.

A good way to check entire oxygen system for a leak is to have systems charged to 450 pounds the night before take-off. If the pressure is less than 400 pounds the next morning a leak is present.

NOTE:

A 50 pound drop in pressure can be anticipated due to cooling off of oxygen after the system is charged.

GENERAL RULES FOR SAFE HANDLING OF OXYGEN

DON'T use oil with oxygen. The reaction when these two substances come in contact is more violent than igniting dynamite. Keep oil or grease away from oxygen cylinders, cylinder valves, and other equipment. Clean hands of oil or grease before using oxygen apparatus.

DON'T wear greasy clothes, gloves, etc., when working with oxygen. Oxygen will cause substances with merely a trace of oil or grease thereon to burn with great intensity.

DON'T use inflammable substances near oxygen. Oxygen itself will not burn but will greatly accelerate combustion.

PASSING OUT AT ALTITUDE

There are two ways to use oxygen equipment at altitude -- a right and wrong. Don't be a fish out of water when you fly above 18,000 feet. Learn how to use oxygen the right way NOW - then you won't have to re-learn the hard way after an unpleasant experience or two.

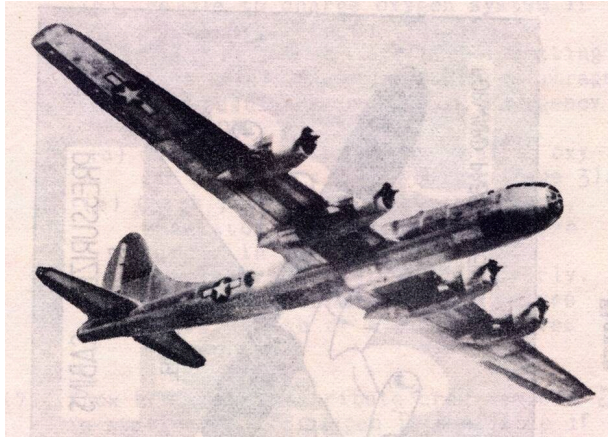
A few points on reviving a fellow crew member who has passed out due to oxygen lack--an opportunity that may well be yours some day!

1. KEEP CALM -- Just because a man passes out from oxygen lack is no sign that he's at death's door. TAKE YOUR TIME. Remember you're

- going to need OXYGEN yourself for the job at hand. If you don't heed this, there'll be TWO to revive instead of only one.
2. PLAN how you're going to revive the victim--then proceed with the least possible exertion to yourself.
 3. DON'T attempt to drag or carry the victim to a supply of oxygen. If he's not near a regulator hose, take portable oxygen TO him. Take plenty -- 3 or 5 bottles -- some for him, plenty for you. If he's near a regulator hose you're in luck. Connect his mask and NOW use the "Emergency." Ordinarily you'll almost blow him right back onto his feet.
 4. GET ASSISTANCE. Have someone keep an eye on you, and supply you with refilled walk-around bottles if needed.
 5. Connect the victim to a walk-around bottle. Ordinarily he'll come to in a matter of seconds. But watch him. He'll be punch drunk and confused as to what's going on.
 6. After he's emptied one bottle, give him another. Then put him on a third and assist him ON HIS OWN POWER to a regulator. Turn the Auto-Mix "OFF" and let him breathe pure oxygen for 5 or 10 minutes. Ordinarily he'll be able to return to his regular duties.
 7. If the victim has been out for a number of minutes he may present more of a problem. Breathing becomes shallow -- and the demand system gives just what is asked of it -- a weak inspiration brings a small squirt of oxygen. So, you'll have to help him with artificial respiration, learn it -- your flight surgeon will show you how.

Caution: Above 20,000 feet, get assistance to give artificial respiration. It will be strenuous work -- a job for several men, not one. Take turns of one to three minutes each.

THE B-29 AT HIGH ALTITUDE



The B-29 is designed to permit you to fly at high altitude without the usual disadvantages of extreme cold, diminished pressure, and decreased oxygen. This is done by the use of pressurized, heated compartments. A highly efficient oxygen system is also incorporated in the ship for use when the cabins are unpressurized.

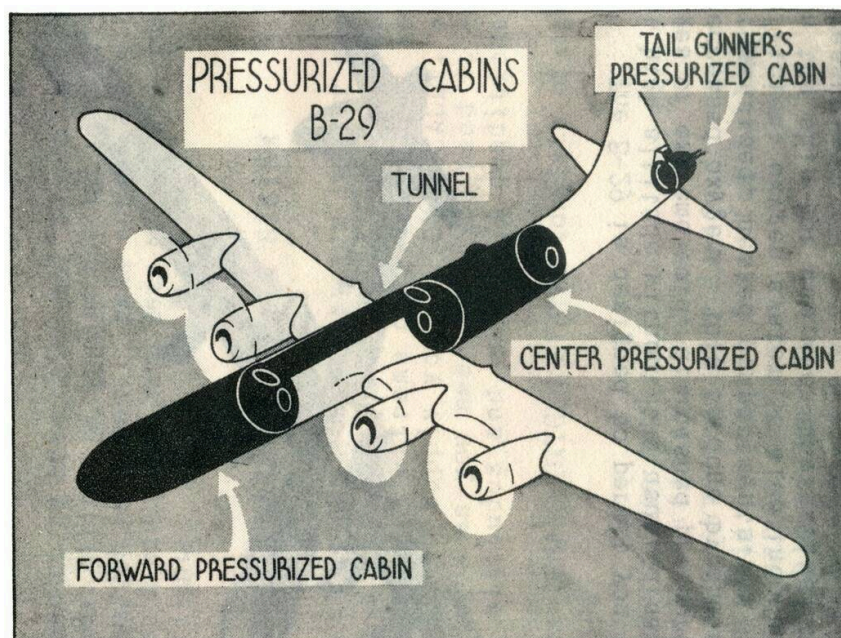


Fig. 13

PRESSURE CABINS

There are three pressure compartments in your ship: Forward, Aft, and Tail. (See Fig. 13). The forward and aft cabins are connected by a

communicating tunnel running over the bomb-bay; the tail compartment is isolated when pressurized.

OPERATION: The cabins are pressurized from the superchargers of the two inboard engines. Your flight engineer controls the air-flow from these superchargers, permitting the cabin Pressure Regulators to automatically regulate the air pressure within the cabin. From ground level to 8,000 ft the cabins are not pressurized. When the cabins are under pressure, your cabin altitude will remain at 8,000 ft, while the ship is flying at any altitude up to 30,000 ft. When your ship gets about 30,000 ft, the cabin pressure will rise proportionately so that it reaches about 12,000 ft when the ship is at 40,000 ft. (See Fig. 14 for this pressure relation). Cabin temperature is controlled by thermostat.

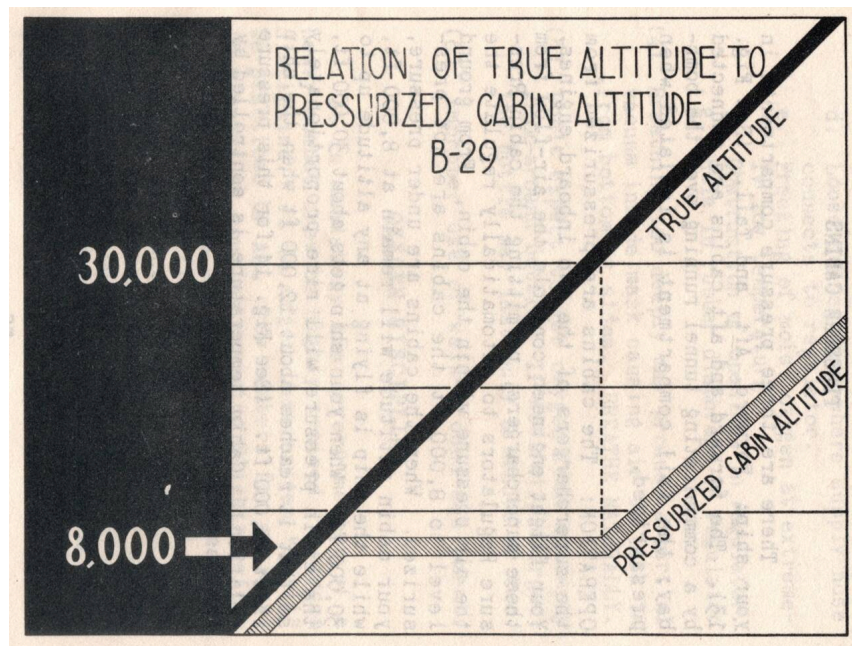


Fig. 14

EMERGENCY PRESSURE RELEASE. Since all emergency doors open in, cabin pressure must be released in all emergencies. This is done by a valve controlled by cable from:

1. Pilot's seat
2. Behind R.H. side gunner

WHEN DO YOU NEED OXYGEN IN A B-29?

As long as the cabin altitude is below 10,000 ft you're O.K. without oxygen.

YOU MUST USE OXYGEN when the **CABIN ALTITUDE** goes **ABOVE** 10,000 feet.

This may occur:

1. When you are flying unpressurized.
2. If you fly above 35,000 feet.
3. Following explosive decompression (Blown blister).
4. In all emergencies above 10,000 feet (Your Pressure must be released so emergency exits can be used).

OXYGEN SYSTEM

GENERAL:

Your oxygen system is a low pressure demand system with 14 oxygen stations corresponding to the various crew positions. The system utilizes the same type of equipment described for the Heavy Bombers:

1. A-14 Demand Mask (See Pg. 8-16).
2. A-12 Demand Regulator (See Pg. 17-19).
3. Pressure Indicator (Pg. 31).
4. Flow Indicator (Pg. 31).
5. Eighteen Type G-1 low pressure cylinders (Pg.37).

DISTRIBUTION:

In the earlier B-29's, the oxygen cylinders were located in the center wing section. At present, however, the cylinders are distributed through the aft-portion of the ship. Regardless of the location of cylinders, distribution is essentially the same. This distribution is highly efficient. The entire loss of the O₂ supply to several crew positions has been practically eliminated by the following principle.

1. EACH OF THE FOURTEEN OXYGEN STATIONS IS SUPPLIED FROM TWO WIDELY SEPARATED DISTRIBUTION LINES DRAWING FROM AT LEAST TWO CYLINDERS EACH. LOSS OF

- ONE LINE OR ITS ASSOCIATED CYLINDERS STILL LEAVES EACH STATION WITH AN ALTERNATE SOURCE OF OXYGEN.
2. The entire system is self-equalizing by use of cross feeds and automatic check valves. If part of the system is shot out, all functioning stations will have equal access to the remaining supply.

Brief study of the diagram of the oxygen supply to a typical B-29 station (Fig. 15) will demonstrate the safety factors of this dual source supply.

Exact location of every oxygen cylinder and supply line may be found in the late tech orders.

PORTABLE EQUIPMENT

Two types of walk-around bottles are furnished in the B-29:

1. SEVEN D-2 TYPE WALK-AROUNDS (See Pg. 25, 26). ONE AT EACH OF FOLLOWING POSITIONS: PILOT, CO-PILOT, ENGINEER, NAVIGATOR, UPPER AND RIGHT GUNNERS AND RADAR OPERATOR.
2. A-4 WALK-AROUND (See Pg. 24) One at each remaining station.

SERVICING:

See Page 47 for servicing instructions. The filler valve is located on the left side of the fuselage.

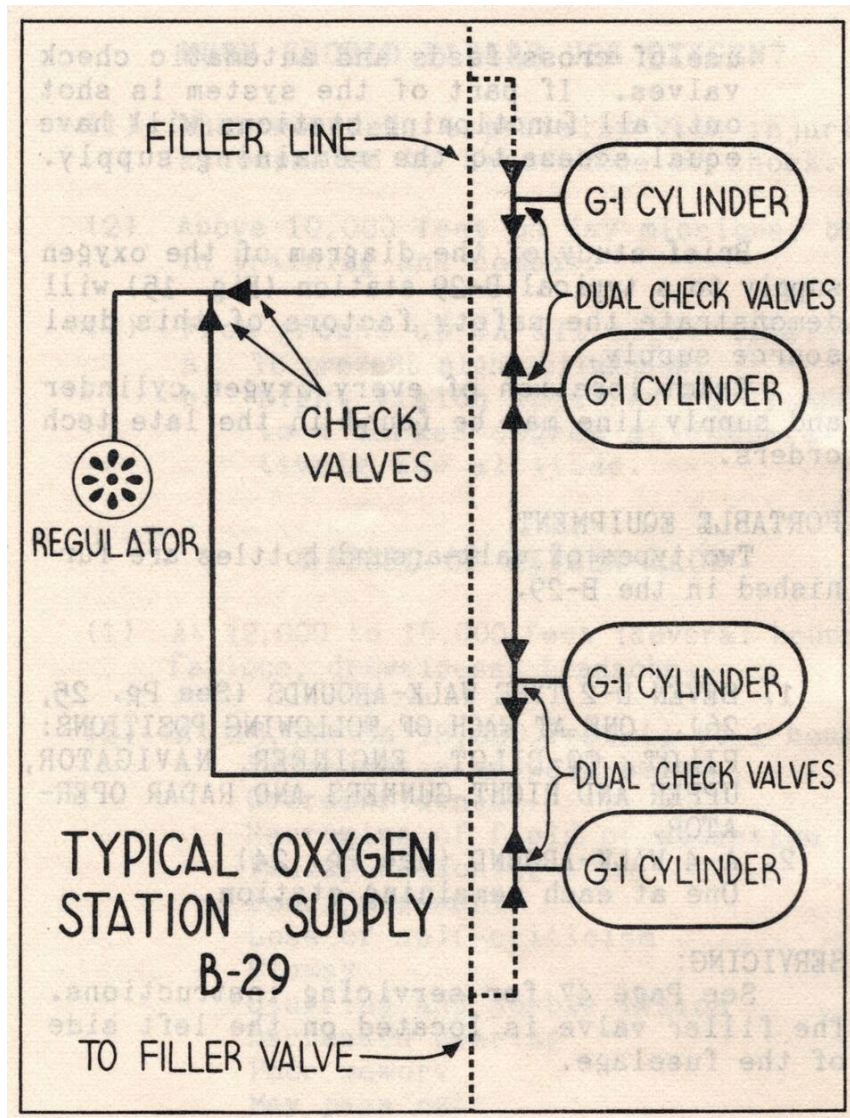


Fig. 15

EXPLOSIVE DECOMPRESSION

With the use of pressurized compartments there is a possibility of explosive decompression. This is most likely to occur if a blister is blown either due to gunfire or a defective blister. Due to the excess capacity of the superchargers, small cabin holes will not cause explosive decompression. If explosive decompression occurred at 30,000 ft, the pressure altitude of the cabins would change from 8,000 ft to 30,000 ft in less than one second. What would happen in such a case?

PHYSICAL EFFECTS:

The human body is affected very little by explosive decompression. IF YOU GET OXYGEN YOU WILL SUFFER NO HARMFUL EFFECT. You may feel a little distension of the belly or a little rush of air from the lungs, but this is generally slight and of no consequence. Your ears should clear automatically since you are going up, not down. The only dangers of explosive decompression are as follows:

DANGERS:

1. Tunnel -- During an explosive decompression a high velocity wind passes through the tunnel sufficiently strong to blow a man out the exits, and cause serious injury. The newer ships will have tunnel doors with small port hole openings; this will eliminate this danger. In ships not equipped in this fashion, **NOBODY SHOULD ENTER THE TUNNEL WHEN THE SHIP IS PRESSURIZED.**
2. Blister -- **ALWAYS FASTEN YOUR SAFETY BELT** at the blister when the ship is pressurized. Otherwise you may be blown out of the ship by an explosive decompression. Also be certain that your mask is secured to your helmet, or it will be blown from the ship.

WHAT TO DO?

The important thing in explosive decompression is to get oxygen rapidly. To get oxygen rapidly you **MUST** at all times:

1. Wear your helmet with the mask attached-let the mask hang from the helmet. Otherwise you may lose your mask and helmet.
2. Keep the regulator hose clipped to your clothing and your mask plugged in.

In case of explosive decompression, adjust your mask to the face and fasten the hook on your helmet. You will then be O.K.

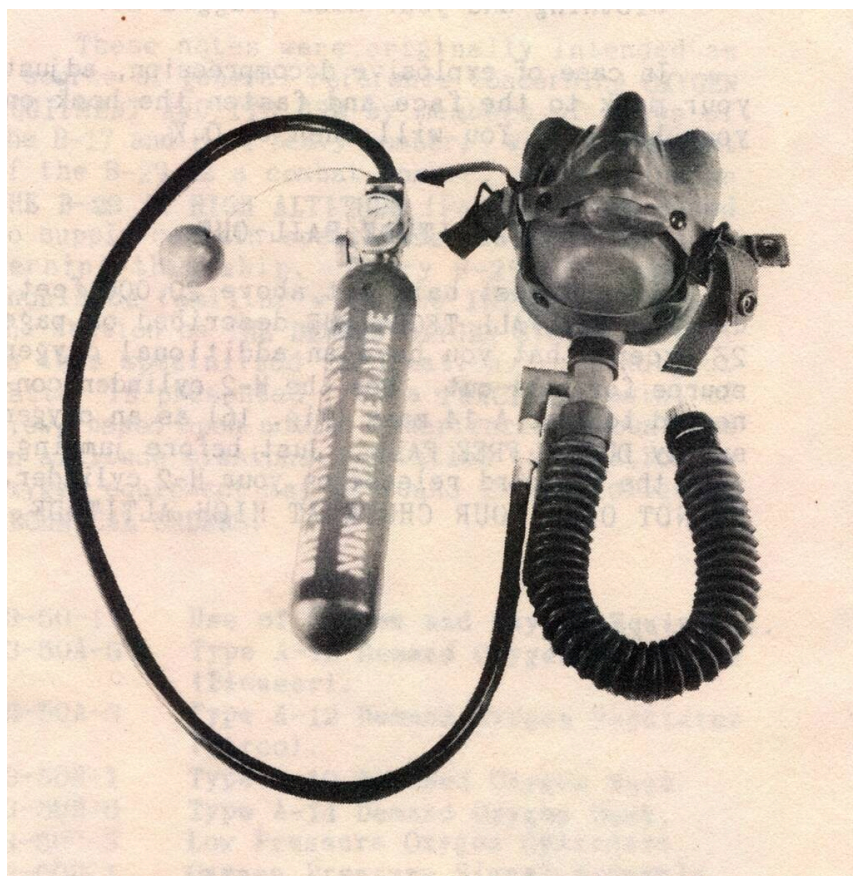


Fig. 16

HIGH ALTITUDE BAIL-OUT

If you must bail out above 20,000 feet, use the FREEFALL TECHNIQUE described on page 23 except that you have an additional oxygen source for bail out. Use the H-2 cylinder connected to your A-14 mask (Fig. 16) as an oxygen supply DURING FREE FALL. Just before jumping, pull the rip cord release on your H-2 cylinder. DO NOT OPEN YOUR CHUTE AT HIGH ALTITUDE.

--REPRODUCED BY--
Reproduction Dept.
Army Air Field Lincoln, Nebr.
10-18-44 20 M

*** END OF THE PROJECT GUTENBERG EBOOK SURVIVAL AT
ALTITUDE FOR HEAVY AND VERY HEAVY BOMBER CREWS ***

Updated editions will replace the previous one—the old editions will be renamed.

Creating the works from print editions not protected by U.S. copyright law means that no one owns a United States copyright in these works, so the Foundation (and you!) can copy and distribute it in the United States without permission and without paying copyright royalties. Special rules, set forth in the General Terms of Use part of this license, apply to copying and distributing Project Gutenberg™ electronic works to protect the PROJECT GUTENBERG™ concept and trademark. Project Gutenberg is a registered trademark, and may not be used if you charge for an eBook, except by following the terms of the trademark license, including paying royalties for use of the Project Gutenberg trademark. If you do not charge anything for copies of this eBook, complying with the trademark license is very easy. You may use this eBook for nearly any purpose such as creation of derivative works, reports, performances and research. Project Gutenberg eBooks may be modified and printed and given away—you may do practically ANYTHING in the United States with eBooks not protected by U.S. copyright law. Redistribution is subject to the trademark license, especially commercial redistribution.

START: FULL LICENSE

THE FULL PROJECT GUTENBERG™ LICENSE

PLEASE READ THIS BEFORE YOU DISTRIBUTE OR USE THIS WORK

To protect the Project Gutenberg™ mission of promoting the free distribution of electronic works, by using or distributing this work (or any other work associated in any way with the phrase “Project Gutenberg”), you agree to comply with all the terms of the Full Project Gutenberg License available with this file or online at www.gutenberg.org/license.

Section 1. General Terms of Use and Redistributing Project Gutenberg electronic works

1.A. By reading or using any part of this Project Gutenberg electronic work, you indicate that you have read, understand, agree to and accept all the terms of this license and intellectual property (trademark/copyright) agreement. If you do not agree to abide by all the terms of this agreement, you must cease using and return or destroy all copies of Project Gutenberg electronic works in your possession. If you paid a fee for obtaining a copy of or access to a Project Gutenberg electronic work and you do not agree to be bound by the terms of this agreement, you may obtain a refund from the person or entity to whom you paid the fee as set forth in paragraph 1.E.8.

1.B. “Project Gutenberg” is a registered trademark. It may only be used on or associated in any way with an electronic work by people who agree to be bound by the terms of this agreement. There are a few things that you can do with most Project Gutenberg electronic works even without complying with the full terms of this agreement. See paragraph 1.C below. There are a lot of things you can do with Project Gutenberg electronic works if you follow the terms of this agreement and help preserve free future access to Project Gutenberg electronic works. See paragraph 1.E below.

1.C. The Project Gutenberg Literary Archive Foundation (“the Foundation” or PGLAF), owns a compilation copyright in the collection of Project Gutenberg electronic works. Nearly all the individual works in the collection are in the public domain in the United States. If an individual work is unprotected by copyright law in the United States and you are

located in the United States, we do not claim a right to prevent you from copying, distributing, performing, displaying or creating derivative works based on the work as long as all references to Project Gutenberg are removed. Of course, we hope that you will support the Project Gutenberg mission of promoting free access to electronic works by freely sharing Project Gutenberg works in compliance with the terms of this agreement for keeping the Project Gutenberg name associated with the work. You can easily comply with the terms of this agreement by keeping this work in the same format with its attached full Project Gutenberg License when you share it without charge with others.

1.D. The copyright laws of the place where you are located also govern what you can do with this work. Copyright laws in most countries are in a constant state of change. If you are outside the United States, check the laws of your country in addition to the terms of this agreement before downloading, copying, displaying, performing, distributing or creating derivative works based on this work or any other Project Gutenberg work. The Foundation makes no representations concerning the copyright status of any work in any country other than the United States.

1.E. Unless you have removed all references to Project Gutenberg:

1.E.1. The following sentence, with active links to, or other immediate access to, the full Project Gutenberg License must appear prominently whenever any copy of a Project Gutenberg work (any work on which the phrase “Project Gutenberg” appears, or with which the phrase “Project Gutenberg” is associated) is accessed, displayed, performed, viewed, copied or distributed:

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg™ License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

1.E.2. If an individual Project Gutenberg electronic work is derived from texts not protected by U.S. copyright law (does not contain a notice indicating that it is posted with permission of the copyright holder), the work can be copied and distributed to anyone in the United States without paying any fees or charges. If you are redistributing or providing access to a work with the phrase “Project Gutenberg” associated with or appearing on the work, you must comply either with the requirements of paragraphs 1.E.1 through 1.E.7 or obtain permission for the use of the work and the Project Gutenberg trademark as set forth in paragraphs 1.E.8 or 1.E.9.

1.E.3. If an individual Project Gutenberg electronic work is posted with the permission of the copyright holder, your use and distribution must comply with both paragraphs 1.E.1 through 1.E.7 and any additional terms imposed by the copyright holder. Additional terms will be linked to the Project Gutenberg License for all works posted with the permission of the copyright holder found at the beginning of this work.

1.E.4. Do not unlink or detach or remove the full Project Gutenberg License terms from this work, or any files containing a part of this work or any other work associated with Project Gutenberg.

1.E.5. Do not copy, display, perform, distribute or redistribute this electronic work, or any part of this electronic work, without prominently displaying the sentence set forth in paragraph 1.E.1 with active links or immediate access to the full terms of the Project Gutenberg License.

1.E.6. You may convert to and distribute this work in any binary, compressed, marked up, nonproprietary or proprietary form, including any word processing or hypertext form. However, if you provide access to or distribute copies of a Project Gutenberg work in a format other than “Plain Vanilla ASCII” or other format used in the official version posted on the official Project Gutenberg website (www.gutenberg.org), you must, at no additional cost, fee or expense to the user, provide a copy, a means of exporting a copy, or a means of obtaining a copy upon request, of the work in its original “Plain Vanilla ASCII” or other form. Any alternate format must include the full Project Gutenberg License as specified in paragraph 1.E.1.

1.E.7. Do not charge a fee for access to, viewing, displaying, performing, copying or distributing any Project Gutenberg works unless you comply with paragraph 1.E.8 or 1.E.9.

1.E.8. You may charge a reasonable fee for copies of or providing access to or distributing Project Gutenberg electronic works provided that:

- You pay a royalty fee of 20% of the gross profits you derive from the use of Project Gutenberg works calculated using the method you already use to calculate your applicable taxes. The fee is owed to the owner of the Project Gutenberg trademark, but he has agreed to donate royalties under this paragraph to the Project Gutenberg Literary Archive Foundation. Royalty payments must be paid within 60 days following each date on which you prepare (or are legally required to prepare) your periodic tax returns. Royalty payments should be clearly marked as such and sent to the Project Gutenberg Literary Archive Foundation at the address specified in Section 4, "Information about donations to the Project Gutenberg Literary Archive Foundation."
- You provide a full refund of any money paid by a user who notifies you in writing (or by e-mail) within 30 days of receipt that s/he does not agree to the terms of the full Project Gutenberg™ License. You must require such a user to return or destroy all copies of the works possessed in a physical medium and discontinue all use of and all access to other copies of Project Gutenberg™ works.
- You provide, in accordance with paragraph 1.F.3, a full refund of any money paid for a work or a replacement copy, if a defect in the electronic work is discovered and reported to you within 90 days of receipt of the work.
- You comply with all other terms of this agreement for free distribution of Project Gutenberg™ works.

1.E.9. If you wish to charge a fee or distribute a Project Gutenberg™ electronic work or group of works on different terms than are set forth in this agreement, you must obtain permission in writing from the Project Gutenberg Literary Archive Foundation, the manager of the Project

Gutenberg™ trademark. Contact the Foundation as set forth in Section 3 below.

1.F.

1.F.1. Project Gutenberg volunteers and employees expend considerable effort to identify, do copyright research on, transcribe and proofread works not protected by U.S. copyright law in creating the Project Gutenberg™ collection. Despite these efforts, Project Gutenberg™ electronic works, and the medium on which they may be stored, may contain “Defects,” such as, but not limited to, incomplete, inaccurate or corrupt data, transcription errors, a copyright or other intellectual property infringement, a defective or damaged disk or other medium, a computer virus, or computer codes that damage or cannot be read by your equipment.

1.F.2. LIMITED WARRANTY, DISCLAIMER OF DAMAGES - Except for the “Right of Replacement or Refund” described in paragraph 1.F.3, the Project Gutenberg Literary Archive Foundation, the owner of the Project Gutenberg™ trademark, and any other party distributing a Project Gutenberg™ electronic work under this agreement, disclaim all liability to you for damages, costs and expenses, including legal fees. YOU AGREE THAT YOU HAVE NO REMEDIES FOR NEGLIGENCE, STRICT LIABILITY, BREACH OF WARRANTY OR BREACH OF CONTRACT EXCEPT THOSE PROVIDED IN PARAGRAPH 1.F.3. YOU AGREE THAT THE FOUNDATION, THE TRADEMARK OWNER, AND ANY DISTRIBUTOR UNDER THIS AGREEMENT WILL NOT BE LIABLE TO YOU FOR ACTUAL, DIRECT, INDIRECT, CONSEQUENTIAL, PUNITIVE OR INCIDENTAL DAMAGES EVEN IF YOU GIVE NOTICE OF THE POSSIBILITY OF SUCH DAMAGE.

1.F.3. LIMITED RIGHT OF REPLACEMENT OR REFUND - If you discover a defect in this electronic work within 90 days of receiving it, you can receive a refund of the money (if any) you paid for it by sending a written explanation to the person you received the work from. If you received the work on a physical medium, you must return the medium with your written explanation. The person or entity that provided you with the defective work may elect to provide a replacement copy in lieu of a refund. If you received the work electronically, the person or entity providing it to

you may choose to give you a second opportunity to receive the work electronically in lieu of a refund. If the second copy is also defective, you may demand a refund in writing without further opportunities to fix the problem.

1.F.4. Except for the limited right of replacement or refund set forth in paragraph 1.F.3, this work is provided to you 'AS-IS', WITH NO OTHER WARRANTIES OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO WARRANTIES OF MERCHANTABILITY OR FITNESS FOR ANY PURPOSE.

1.F.5. Some states do not allow disclaimers of certain implied warranties or the exclusion or limitation of certain types of damages. If any disclaimer or limitation set forth in this agreement violates the law of the state applicable to this agreement, the agreement shall be interpreted to make the maximum disclaimer or limitation permitted by the applicable state law. The invalidity or unenforceability of any provision of this agreement shall not void the remaining provisions.

1.F.6. INDEMNITY - You agree to indemnify and hold the Foundation, the trademark owner, any agent or employee of the Foundation, anyone providing copies of Project Gutenberg™ electronic works in accordance with this agreement, and any volunteers associated with the production, promotion and distribution of Project Gutenberg™ electronic works, harmless from all liability, costs and expenses, including legal fees, that arise directly or indirectly from any of the following which you do or cause to occur: (a) distribution of this or any Project Gutenberg work, (b) alteration, modification, or additions or deletions to any Project Gutenberg work, and (c) any Defect you cause.

Section 2. Information about the Mission of Project Gutenberg

Project Gutenberg is synonymous with the free distribution of electronic works in formats readable by the widest variety of computers including obsolete, old, middle-aged and new computers. It exists because of the

efforts of hundreds of volunteers and donations from people in all walks of life.

Volunteers and financial support to provide volunteers with the assistance they need are critical to reaching Project Gutenberg's goals and ensuring that the Project Gutenberg collection will remain freely available for generations to come. In 2001, the Project Gutenberg Literary Archive Foundation was created to provide a secure and permanent future for Project Gutenberg and future generations. To learn more about the Project Gutenberg Literary Archive Foundation and how your efforts and donations can help, see Sections 3 and 4 and the Foundation information page at www.gutenberg.org.

Section 3. Information about the Project Gutenberg Literary Archive Foundation

The Project Gutenberg Literary Archive Foundation is a non-profit 501(c)(3) educational corporation organized under the laws of the state of Mississippi and granted tax exempt status by the Internal Revenue Service. The Foundation's EIN or federal tax identification number is 64-6221541. Contributions to the Project Gutenberg Literary Archive Foundation are tax deductible to the full extent permitted by U.S. federal laws and your state's laws.

The Foundation's business office is located at 41 Watchung Plaza #516, Montclair NJ 07042, USA, +1 (862) 621-9288. Email contact links and up to date contact information can be found at the Foundation's website and official page at www.gutenberg.org/contact

Section 4. Information about Donations to the Project Gutenberg Literary Archive Foundation

Project Gutenberg™ depends upon and cannot survive without widespread public support and donations to carry out its mission of increasing the number of public domain and licensed works that can be freely distributed in machine-readable form accessible by the widest array of equipment

including outdated equipment. Many small donations (\$1 to \$5,000) are particularly important to maintaining tax exempt status with the IRS.

The Foundation is committed to complying with the laws regulating charities and charitable donations in all 50 states of the United States. Compliance requirements are not uniform and it takes a considerable effort, much paperwork and many fees to meet and keep up with these requirements. We do not solicit donations in locations where we have not received written confirmation of compliance. To SEND DONATIONS or determine the status of compliance for any particular state visit www.gutenberg.org/donate.

While we cannot and do not solicit contributions from states where we have not met the solicitation requirements, we know of no prohibition against accepting unsolicited donations from donors in such states who approach us with offers to donate.

International donations are gratefully accepted, but we cannot make any statements concerning tax treatment of donations received from outside the United States. U.S. laws alone swamp our small staff.

Please check the Project Gutenberg web pages for current donation methods and addresses. Donations are accepted in a number of other ways including checks, online payments and credit card donations. To donate, please visit: www.gutenberg.org/donate.

Section 5. General Information About Project Gutenberg electronic works

Professor Michael S. Hart was the originator of the Project Gutenberg concept of a library of electronic works that could be freely shared with anyone. For forty years, he produced and distributed Project Gutenberg eBooks with only a loose network of volunteer support.

Project Gutenberg eBooks are often created from several printed editions, all of which are confirmed as not protected by copyright in the U.S. unless a

copyright notice is included. Thus, we do not necessarily keep eBooks in compliance with any particular paper edition.

Most people start at our website which has the main PG search facility:
www.gutenberg.org.

This website includes information about Project Gutenberg, including how to make donations to the Project Gutenberg Literary Archive Foundation, how to help produce our new eBooks, and how to subscribe to our email newsletter to hear about new eBooks.